

Tungsten Carbide End Mills UNIMAX Series

Vol. 2

HARDMAX 4 Flutes Long Neck Radius End Mills

Total 381 Models

Add 42

HLRS4000

HARDMAX 2 Flutes Long Neck Radius End Mills

Total 353 Models

HLRS2000/2000E



UNION TOOL CO.

HARDMAX 4 Flute Long Neck Radius End Mills

Size $\varnothing 0.2 \sim \varnothing 6$



HLRS4000

Super
MG

HARD
MAX

30°

R

± 0.005

Shank Dia
 $\varnothing \pm 0.005$

Back Taper
Geometry

Variable
Pitch

Additional 42 Models

Material Applications (☆ Highly Recommended ◎ Recommended ○ Suggested)

Work Material														
CARBON STEELS S45C S55C	ALLOY STEELS SK / SCM SUS	PREHARDENED STEELS NAK HPM	HARDENED STEELS			CAST IRON	ALUMINUM ALLOYS	GRAPHITE	COPPER	PLASTICS	GLASS FILLED PLASTICS	TITANIUM ALLOYS	HEAT RESISTANT ALLOYS	CEMENTED CARBIDE
			~55HRC	~60HRC	~70HRC									
○	○	◎	◎	◎	◎ ~65HRC	○			○			○	○	

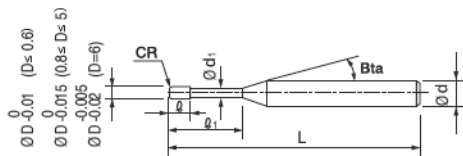
Features

Long Neck Radius design for high efficiency and high quality milling.

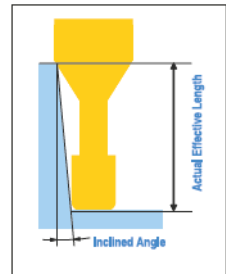
The rigid tool geometry offers longer tool life when milling Hard Materials.

HARDMAX coat is adopted which maintains heat resistance, toughness and lubricity at a high level.

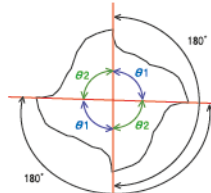
Both dry and wet coolant offer stable and long tool life.



The shank taper angle shown is not an exact value and to avoid contact with the work piece, we recommend the user controls the precise value of this angle. Shank taper angle should not make contact with the work piece.

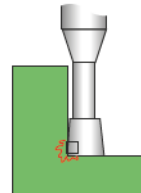


Feature 1 : Variable pitch



$\varnothing 1 > \varnothing 2$: The unequal division reduces chattering and tip damage.
 $\varnothing 1 + \varnothing 2 = 180^\circ$: Easy to measure outside diameter.

Feature 2 : Back taper geometry



The back taper geometry reduces cutting resistance, which enables stable milling on vertical walls.

Total 381 models

Unit (mm)

Model Number	Outside Diameter ØD	Corner Radius CR	Effective Length ℓ_1	Length of Cut ℓ	Neck Diameter Ød1	Shank Taper Angle Bta	Overall Length L	Shank Diameter Ød	Price ¥	Effective Length by Inclined Angles				
										30°	1°	1°30'	2°	3°
HLRS 4002-002-003	0.2	R0.02	0.3	0.12	0.185	16°	50	4	14,100	0.39	0.41	0.44	0.46	0.51
HLRS 4002-002-005			0.5				50	4	14,100	0.60	0.63	0.67	0.70	0.75
HLRS 4002-002-010			1				50	4	14,100	1.13	1.18	1.22	1.27	1.36
HLRS 4002-002-015			1.5				50	4	16,590	1.65	1.71	1.77	1.84	1.98
HLRS 4002-002-020		R0.05	2				50	4	18,150	2.17	2.25	2.32	2.41	2.59
HLRS 4002-005-003			0.3				50	4	14,100	0.39	0.41	0.43	0.46	0.50

* Additional model

HARDMAX 4 Flute Long Neck Radius End Mills

Model Number	Outside Diameter Ø D	Corner Radius CR	Effective Length ℓ_1	Length of Cut ℓ	Neck Diameter Ø d ₁	Shank Taper Angle Bta	Overall Length L	Shank Diameter Ø d	Price ¥	Effective Length by Inclined Angles				
										30°	1°	1°30'	2°	3°
* HLRS 4003-002-003	0.3	R0.02	0.3	0.18	0.28	16°	50	4	14,100	0.41	0.43	0.46	0.48	0.52
HLRS 4003-002-005			0.5				50	4	14,100	0.62	0.65	0.68	0.71	0.77
HLRS 4003-002-010			1				50	4	14,100	1.15	1.19	1.24	1.28	1.38
* HLRS 4003-002-015			1.5				50	4	14,100	1.67	1.73	1.79	1.85	1.99
* HLRS 4003-002-020			2				50	4	14,100	2.19	2.26	2.34	2.42	2.60
* HLRS 4003-005-003		R0.05	0.3				50	4	14,100	0.41	0.43	0.45	0.47	0.52
HLRS 4003-005-005	0.4	R0.02	0.5	0.24	0.385	16°	50	4	14,100	0.62	0.65	0.68	0.71	0.76
HLRS 4004-002-005			1				50	4	9,050	0.62	0.65	0.68	0.71	0.77
HLRS 4004-002-010			2				50	4	9,050	1.15	1.19	1.24	1.28	1.38
HLRS 4004-002-020			5				50	4	9,050	2.17	2.25	2.33	2.41	2.59
HLRS 4004-005-005		R0.05	0.5				50	4	9,050	0.62	0.65	0.68	0.71	0.76
HLRS 4004-005-010			1				50	4	9,050	1.14	1.19	1.23	1.28	1.37
HLRS 4004-005-015			1.5				50	4	9,050	1.65	1.72	1.77	1.84	1.97
HLRS 4004-005-020			2				50	4	9,050	2.17	2.25	2.32	2.41	2.59
HLRS 4004-01-010		R0.1	1				50	4	9,050	1.14	1.19	1.23	1.27	1.36
HLRS 4004-01-020			2				50	4	9,050	2.17	2.25	2.32	2.40	2.57
HLRS 4005-002-010	0.5	R0.02	1	0.3	0.485	16°	50	4	7,370	1.15	1.19	1.24	1.28	1.38
HLRS 4005-002-020			2				50	4	7,370	2.17	2.25	2.33	2.41	2.59
HLRS 4005-005-010		R0.05	1				50	4	7,370	1.14	1.19	1.23	1.28	1.37
HLRS 4005-005-020			2				50	4	7,370	2.17	2.25	2.32	2.41	2.59
HLRS 4005-01-010		R0.1	1				50	4	7,370	1.14	1.19	1.23	1.27	1.36
HLRS 4005-01-020			2				50	4	7,370	2.17	2.25	2.32	2.40	2.57
HLRS 4006-005-020	0.6	R0.05	2	0.36	0.585	16°	50	4	7,370	2.17	2.25	2.32	2.41	2.59
HLRS 4006-005-040			4				50	4	7,370	4.24	4.38	4.53	4.68	5.03
HLRS 4006-01-020		R0.1	2				50	4	7,370	2.17	2.25	2.32	2.40	2.57
HLRS 4006-01-040			4				50	4	7,370	4.24	4.38	4.52	4.68	5.02
HLRS 4008-002-020	0.8	R0.02	2	0.48	0.78	16°	50	4	8,100	2.51	2.70	2.87	3.02	3.29
HLRS 4008-002-030			3				50	4	8,100	3.59	3.82	4.02	4.19	4.51
HLRS 4008-002-040			4				50	4	8,400	4.66	4.92	5.14	5.34	5.74
HLRS 4008-002-060			6				50	4	8,400	6.78	7.10	7.36	7.61	8.19
HLRS 4008-002-080			8				50	4	8,400	8.88	9.25	9.56	9.89	10.63
HLRS 4008-005-020		R0.05	2				50	4	8,100	2.51	2.70	2.86	3.01	3.28
HLRS 4008-005-030			3				50	4	8,100	3.59	3.82	4.01	4.19	4.51
HLRS 4008-005-040			4				50	4	8,400	4.66	4.92	5.14	5.33	5.73
HLRS 4008-005-060			6				50	4	8,400	6.77	7.09	7.35	7.61	8.18
HLRS 4008-01-020		R0.1	2				50	4	8,100	2.51	2.69	2.86	3.00	3.27
HLRS 4008-01-030			3				50	4	8,100	3.58	3.81	4.01	4.18	4.50
HLRS 4008-01-040			4				50	4	8,400	4.65	4.92	5.13	5.33	5.72
HLRS 4008-01-060			6				50	4	8,400	6.77	7.09	7.35	7.60	8.17
HLRS 4008-02-020		R0.2	2				50	4	8,100	2.49	2.68	2.84	2.98	3.25
HLRS 4008-02-030			3				50	4	8,100	3.57	3.80	3.99	4.16	4.47
HLRS 4008-02-040			4				50	4	8,400	4.64	4.90	5.12	5.31	5.70
HLRS 4008-02-060			6				50	4	8,400	6.76	7.08	7.34	7.59	8.14
HLRS 4010-002-020	1	R0.02	2	0.8	0.95	16°	50	4	7,400	2.57	2.75	2.90	3.05	3.32
HLRS 4010-002-030			3				50	4	7,400	3.64	3.86	4.05	4.22	4.54
HLRS 4010-002-040			4				50	4	7,400	4.70	4.96	5.17	5.36	5.76
HLRS 4010-002-050			5				50	4	8,100	5.76	6.04	6.28	6.50	6.99
HLRS 4010-002-060			6				50	4	8,100	6.81	7.12	7.38	7.64	8.21
HLRS 4010-002-080			8				50	4	8,100	8.91	9.27	9.58	9.92	10.66
HLRS 4010-002-100			10				50	4	8,100	11.00	11.40	11.78	12.19	13.11
* HLRS 4010-002-120			12				55	4	8,100	13.08	13.53	13.98	14.47	15.55

* Additional model

HARDMAX 4 Flute Long Neck Radius End Mills

Model Number	Outside Diameter ØD	Corner Radius CR	Effective Length ℓ_1	Length of Cut ℓ	Neck Diameter Ød ₁	Shank Taper Angle Eta	Overall Length L	Shank Diameter Ød	Price ¥	Effective Length by Inclined Angles				
										30°	1°	1°30'	2°	3°
HLRS 4010-005-020	1	R0.05	2	0.8	0.95	16°	50	4	7,400	2.56	2.74	2.90	3.05	3.31
HLRS 4010-005-030			3				50	4	7,400	3.63	3.85	4.04	4.21	4.53
HLRS 4010-005-040			4				50	4	7,400	4.70	4.95	5.17	5.36	5.76
HLRS 4010-005-050			5				50	4	8,100	5.76	6.04	6.28	6.50	6.98
HLRS 4010-005-060			6				50	4	8,100	6.81	7.12	7.38	7.63	8.20
HLRS 4010-005-080			8				50	4	8,100	8.91	9.27	9.58	9.91	10.65
HLRS 4010-005-100			10				50	4	8,100	11.00	11.40	11.78	12.19	13.10
HLRS 4010-005-120			12				55	4	8,100	13.08	13.53	13.98	14.47	15.55
HLRS 4010-01-020		R0.1	2				50	4	7,400	2.56	2.74	2.89	3.04	3.30
HLRS 4010-01-030			3				50	4	7,400	3.63	3.85	4.04	4.20	4.52
HLRS 4010-01-040			4				50	4	7,400	4.70	4.95	5.16	5.35	5.75
HLRS 4010-01-050			5				50	4	8,100	5.75	6.04	6.27	6.49	6.97
HLRS 4010-01-060			6				50	4	8,100	6.81	7.12	7.37	7.63	8.19
HLRS 4010-01-080			8				50	4	8,100	8.91	9.26	9.57	9.90	10.64
HLRS 4010-01-100			10				50	4	8,100	11.00	11.39	11.77	12.18	13.09
HLRS 4010-01-120			12				55	4	8,100	13.08	13.52	13.98	14.46	15.54
HLRS 4010-02-020		R0.2	2				50	4	7,400	2.55	2.72	2.88	3.02	3.28
HLRS 4010-02-030			3				50	4	7,400	3.62	3.84	4.02	4.19	4.50
HLRS 4010-02-040			4				50	4	7,400	4.69	4.94	5.15	5.34	5.72
HLRS 4010-02-050			5				50	4	8,100	5.75	6.03	6.26	6.47	6.95
HLRS 4010-02-060			6				50	4	8,100	6.80	7.11	7.36	7.61	8.17
HLRS 4010-02-080			8				50	4	8,100	8.90	9.26	9.56	9.89	10.62
HLRS 4010-02-100			10				50	4	8,100	10.99	11.39	11.76	12.17	13.07
HLRS 4010-02-120			12				55	4	8,100	13.07	13.52	13.97	14.45	15.51
HLRS 4010-03-020		R0.3	2				50	4	7,400	2.54	2.71	2.86	3.00	3.25
HLRS 4010-03-030			3				50	4	7,400	3.62	3.83	4.01	4.17	4.48
HLRS 4010-03-040			4				50	4	7,400	4.68	4.93	5.14	5.32	5.70
HLRS 4010-03-050			5				50	4	8,100	5.74	6.02	6.25	6.46	6.93
HLRS 4010-03-060			6				50	4	8,100	6.80	7.10	7.35	7.60	8.15
HLRS 4010-03-080			8				50	4	8,100	8.90	9.25	9.55	9.88	10.60
HLRS 4010-03-100			10				50	4	8,100	10.99	11.38	11.75	12.15	13.04
HLRS 4010-03-120			12				55	4	8,100	13.07	13.51	13.96	14.43	15.49
HLRS 4010-03-160			16				60	4	10,740	17.22	17.77	18.36	18.99	20.39
HLRS 4012-01-040	1.2	R0.1	4	0.96	1.14	16°	50	4	8,400	4.13	4.27	4.41	4.56	4.90
HLRS 4012-01-060			6				50	4	8,400	6.20	6.40	6.61	6.84	7.34
HLRS 4012-01-100			10				50	4	8,400	10.32	10.66	11.01	11.39	12.24
HLRS 4012-02-040		R0.2	4				50	4	8,400	4.13	4.26	4.40	4.55	4.87
HLRS 4012-02-060			6				50	4	8,400	6.19	6.39	6.60	6.82	7.32
HLRS 4012-02-100			10				50	4	8,400	10.32	10.65	11.00	11.38	12.22
HLRS 4012-03-040		R0.3	4				50	4	8,400	4.13	4.25	4.39	4.53	4.85
HLRS 4012-03-060			6				50	4	8,400	6.19	6.38	6.59	6.81	7.30
HLRS 4012-03-100			10				50	4	8,400	10.32	10.64	10.99	11.36	12.20
HLRS 4015-005-030	1.5	R0.05	3	1.2	1.45	16°	50	4	7,900	3.10	3.20	3.31	3.43	3.68
HLRS 4015-005-040			4				50	4	7,900	4.14	4.27	4.41	4.57	4.91
HLRS 4015-005-060			6				50	4	7,900	6.20	6.40	6.61	6.84	7.35
HLRS 4015-005-080			8				50	4	8,200	8.26	8.53	8.82	9.12	9.80
HLRS 4015-005-120			12				55	4	8,200	12.39	12.79	13.22	13.68	14.70
HLRS 4015-005-160			16				60	4	8,200	16.51	17.05	17.62	18.23	19.59

* Additional model

HARDMAX 4 Flute Long Neck Radius End Mills

Model Number	Outside Diameter ØD	Corner Radius CR	Effective Length ℓ_1	Length of Cut ℓ	Neck Diameter Ød ₁	Shank Taper Angle Eta	Overall Length L	Shank Diameter Ød	Price ¥	Effective Length by Inclined Angles				
										30°	1°	1°30'	2°	3°
HLRS 4015-01-030	1.5	R0.1	3	1.2	1.45	16°	50	4	7,900	3.10	3.20	3.31	3.42	3.67
HLRS 4015-01-040			4				50	4	7,900	4.13	4.27	4.41	4.56	4.90
HLRS 4015-01-060			6				50	4	7,900	6.20	6.40	6.61	6.84	7.34
HLRS 4015-01-080			8				50	4	8,200	8.26	8.53	8.81	9.11	9.79
HLRS 4015-01-100			10				50	4	8,200	10.32	10.66	11.01	11.39	12.24
HLRS 4015-01-120			12				55	4	8,200	12.39	12.79	13.21	13.67	14.69
HLRS 4015-01-160			16				60	4	8,200	16.51	17.04	17.61	18.22	19.58
HLRS 4015-01-180			18				60	4	8,200	18.57	19.17	19.82	20.50	22.03
HLRS 4015-02-030		R0.2	3				50	4	7,900	3.10	3.20	3.30	3.41	3.65
HLRS 4015-02-040			4				50	4	7,900	4.13	4.26	4.40	4.55	4.87
HLRS 4015-02-060			6				50	4	7,900	6.19	6.39	6.60	6.82	7.32
HLRS 4015-02-080			8				50	4	8,200	8.26	8.52	8.80	9.10	9.77
HLRS 4015-02-100			10				50	4	8,200	10.32	10.65	11.00	11.38	12.22
HLRS 4015-02-120			12				55	4	8,200	12.38	12.78	13.20	13.66	14.66
HLRS 4015-02-160			16				60	4	8,200	16.51	17.04	17.60	18.21	19.56
HLRS 4015-02-180			18				60	4	8,200	18.57	19.17	19.81	20.49	22.01
HLRS 4015-03-030		R0.3	3				50	4	7,900	3.10	3.19	3.29	3.39	3.63
HLRS 4015-03-040			4				50	4	7,900	4.13	4.25	4.39	4.53	4.85
HLRS 4015-03-060			6				50	4	7,900	6.19	6.38	6.59	6.81	7.30
HLRS 4015-03-080			8				50	4	8,200	8.25	8.51	8.79	9.09	9.75
HLRS 4015-03-100			10				50	4	8,200	10.32	10.64	10.99	11.36	12.20
HLRS 4015-03-120			12				55	4	8,200	12.38	12.77	13.19	13.64	14.64
HLRS 4015-03-160			16				60	4	8,200	16.50	17.03	17.59	18.20	19.54
HLRS 4015-03-180			18				60	4	8,200	18.57	19.16	19.80	20.47	21.98
HLRS 4015-05-030		R0.5	3				50	4	7,900	3.09	3.17	3.27	3.36	3.58
HLRS 4015-05-040			4				50	4	7,900	4.12	4.24	4.37	4.50	4.81
HLRS 4015-05-060			6				50	4	7,900	6.18	6.37	6.57	6.78	7.25
HLRS 4015-05-080			8				50	4	8,200	8.25	8.50	8.77	9.06	9.70
HLRS 4015-05-100			10				50	4	8,200	10.31	10.63	10.97	11.34	12.15
HLRS 4015-05-120			12				55	4	8,200	12.37	12.76	13.17	13.61	14.60
HLRS 4015-05-160			16				60	4	8,200	16.50	17.02	17.57	18.17	19.49
HLRS 4015-05-180			18				60	4	8,200	18.56	19.15	19.77	20.44	21.94
HLRS 4018-02-080	1.8	R0.2	8	1.44	1.72	16°	50	4	8,200	8.29	8.56	8.84	9.14	9.81
HLRS 4018-02-100			10				50	4	8,200	10.36	10.69	11.04	11.42	12.26
HLRS 4018-02-120			12				55	4	8,200	12.42	12.82	13.24	13.69	14.71
HLRS 4018-02-140			14				55	4	8,200	14.48	14.95	15.44	15.97	17.15
HLRS 4018-02-160			16				60	4	8,200	16.54	17.08	17.64	18.25	19.60
HLRS 4020-002-040	2	R0.02	4	1.6	1.92	16°	50	4	7,900	4.17	4.31	4.45	4.61	4.96
HLRS 4020-002-060			6				50	4	7,900	6.24	6.44	6.66	6.89	7.40
HLRS 4020-002-080			8				50	4	8,200	8.30	8.57	8.86	9.16	9.85
HLRS 4020-002-100			10				50	4	8,200	10.36	10.70	11.06	11.44	12.30
HLRS 4020-002-120			12				55	4	8,200	12.42	12.83	13.26	13.72	14.75
HLRS 4020-002-160			16				60	4	8,200	16.55	17.09	17.66	18.27	No interference
HLRS 4020-002-200			20				60	4	8,200	20.67	21.35	22.06	22.83	No interference
HLRS 4020-005-040		R0.05	4				50	4	7,900	4.17	4.31	4.45	4.61	4.95
HLRS 4020-005-060			6				50	4	7,900	6.23	6.44	6.65	6.88	7.40
HLRS 4020-005-080			8				50	4	8,200	8.30	8.57	8.85	9.16	9.84
HLRS 4020-005-100			10				50	4	8,200	10.36	10.70	11.05	11.44	12.29
HLRS 4020-005-120			12				55	4	8,200	12.42	12.83	13.26	13.72	14.74
HLRS 4020-005-160			16				60	4	8,200	16.55	17.08	17.66	18.27	No interference
HLRS 4020-005-200			20				60	4	8,200	20.67	21.34	22.06	22.82	No interference

* Additional model

HARDMAX 4 Flute Long Neck Radius End Mills

Model Number	Outside Diameter Ø D	Corner Radius CR	Effective Length ℓ_1	Length of Cut ℓ	Neck Diameter Ø d ₁	Shank Taper Angle Eta	Overall Length L	Shank Diameter Ø d	Price ¥	Effective Length by Inclined Angles				
										30°	1°	1°30'	2°	3°
HLRS 4020-01-040	2	R0.1	4	1.6	1.92	16°	50	4	7,900	4.17	4.30	4.45	4.60	4.94
HLRS 4020-01-060			6				50	4	7,900	6.23	6.43	6.65	6.88	7.39
HLRS 4020-01-080			8				50	4	8,200	8.30	8.56	8.85	9.15	9.83
HLRS 4020-01-100			10				50	4	8,200	10.36	10.69	11.05	11.43	12.28
HLRS 4020-01-120			12				55	4	8,200	12.42	12.82	13.25	13.71	14.73
HLRS 4020-01-160			16				60	4	8,200	16.55	17.08	17.65	18.26	No Interference
HLRS 4020-01-200			20				60	4	8,200	20.67	21.34	22.05	22.82	No Interference
HLRS 4020-01-240			24				70	4	8,200	24.80	25.60	26.46	27.37	No Interference
HLRS 4020-02-040		R0.2	4				50	4	7,900	4.17	4.30	4.44	4.59	4.92
HLRS 4020-02-060			6				50	4	7,900	6.23	6.43	6.64	6.86	7.36
HLRS 4020-02-080			8				50	4	8,200	8.29	8.56	8.84	9.14	9.81
HLRS 4020-02-100			10				50	4	8,200	10.36	10.69	11.04	11.42	12.26
HLRS 4020-02-120			12				55	4	8,200	12.42	12.82	13.24	13.69	14.71
HLRS 4020-02-160			16				60	4	8,200	16.54	17.08	17.64	18.25	19.60
HLRS 4020-02-200			20				60	4	8,200	20.67	21.33	22.04	22.80	No Interference
HLRS 4020-02-240			24				70	4	8,200	24.79	25.59	26.45	27.36	No Interference
HLRS 4020-03-040		R0.3	4				50	4	7,900	4.16	4.29	4.43	4.57	4.90
HLRS 4020-03-060			6				50	4	7,900	6.23	6.42	6.63	6.85	7.34
HLRS 4020-03-080			8				50	4	8,200	8.29	8.55	8.83	9.13	9.79
HLRS 4020-03-100			10				50	4	8,200	10.35	10.68	11.03	11.40	12.24
HLRS 4020-03-120			12				55	4	8,200	12.41	12.81	13.23	13.68	14.68
HLRS 4020-03-160			16				60	4	8,200	16.54	17.07	17.63	18.24	19.58
HLRS 4020-03-200			20				60	4	8,200	20.67	21.33	22.03	22.79	No Interference
HLRS 4020-03-240			24				70	4	8,200	24.79	25.59	26.44	27.35	No Interference
HLRS 4020-05-040		R0.5	4				50	4	7,900	4.16	4.28	4.40	4.54	4.85
HLRS 4020-05-060			6				50	4	7,900	6.22	6.41	6.61	6.82	7.30
HLRS 4020-05-080			8				50	4	8,200	8.28	8.54	8.81	9.10	9.74
HLRS 4020-05-100			10				50	4	8,200	10.34	10.67	11.01	11.37	12.19
HLRS 4020-05-120			12				55	4	8,200	12.41	12.79	13.21	13.65	14.64
HLRS 4020-05-160			16				60	4	8,200	16.53	17.05	17.61	18.21	19.53
HLRS 4020-05-200			20				60	4	8,200	20.66	21.31	22.01	22.76	No Interference
HLRS 4020-05-240			24				70	4	8,200	24.78	25.57	26.41	27.32	No Interference
HLRS 4020-05-260			26				70	4	8,200	26.85	27.70	28.62	29.59	No Interference
HLRS 4020-05-300			30				70	4	8,200	30.97	31.96	33.02	No Interference	No Interference
HLRS 4025-01-060	2.5	R0.1	6	2	2.42	16°	50	4	8,600	6.23	6.43	6.65	6.88	7.39
HLRS 4025-01-080			8				50	4	8,600	8.30	8.56	8.85	9.15	9.83
HLRS 4025-01-100			10				50	4	8,600	10.36	10.69	11.05	11.43	12.28
HLRS 4025-01-160			16				60	4	8,900	16.55	17.08	17.65	18.26	No Interference
HLRS 4025-01-200			20				60	4	8,900	20.67	21.34	22.05	No Interference	No Interference
HLRS 4025-01-300			30				70	4	9,200	30.99	31.99	No Interference	No Interference	No Interference
HLRS 4025-02-060		R0.2	6				50	4	8,600	6.23	6.43	6.64	6.86	7.36
HLRS 4025-02-080			8				50	4	8,600	8.29	8.56	8.84	9.14	9.81
HLRS 4025-02-100			10				50	4	8,600	10.36	10.69	11.04	11.42	12.26
HLRS 4025-02-160			16				60	4	8,900	16.54	17.08	17.64	18.25	No Interference
HLRS 4025-02-200			20				60	4	8,900	20.67	21.33	22.04	No Interference	No Interference
HLRS 4025-02-300			30				70	4	9,200	30.98	31.98	No Interference	No Interference	No Interference

* Additional model

HARDMAX 4 Flute Long Neck Radius End Mills

Model Number	Outside Diameter ØD	Corner Radius CR	Effective Length ℓ_1	Length of Cut ℓ	Neck Diameter Ød ₁	Shank Taper Angle Eta	Overall Length L	Shank Diameter Ød	Price ¥	Effective Length by Inclined Angles				
										30°	1°	1°30'	2°	3°
HLRS 4025-03-060	2.5	R0.3	6	2	2.42	16°	50	4	8,600	6.23	6.42	6.63	6.85	7.34
HLRS 4025-03-080			8				50	4	8,600	8.29	8.55	8.83	9.13	9.79
HLRS 4025-03-100			10				50	4	8,600	10.35	10.68	11.03	11.40	12.24
HLRS 4025-03-160			16				60	4	8,900	16.54	17.07	17.63	18.24	No interference
HLRS 4025-03-200			20				60	4	8,900	20.67	21.33	22.03	No interference	No interference
HLRS 4025-03-300			30				70	4	9,200	30.98	31.98	No interference	No interference	No interference
HLRS 4025-05-060		R0.5	6				50	4	8,600	6.22	6.41	6.61	6.82	7.30
HLRS 4025-05-080			8				50	4	8,600	8.28	8.54	8.81	9.10	9.74
HLRS 4025-05-100			10				50	4	8,600	10.34	10.67	11.01	11.37	12.19
HLRS 4025-05-160			16				60	4	8,900	16.53	17.05	17.61	18.21	No interference
HLRS 4025-05-200			20				60	4	8,900	20.66	21.31	22.01	No interference	No interference
HLRS 4025-05-300			30				70	4	9,200	30.97	31.96	No interference	No interference	No interference
HLRS 4030-005-040	3	R0.05	4	2.4	2.92	16°	55	6	7,100	4.17	4.31	4.45	4.60	4.95
HLRS 4030-005-060			6				55	6	7,100	6.23	6.43	6.65	6.88	7.40
HLRS 4030-005-080			8				55	6	7,100	8.30	8.56	8.85	9.16	9.84
HLRS 4030-005-100			10				55	6	7,100	10.36	10.69	11.05	11.44	12.29
HLRS 4030-005-120			12				55	6	8,600	12.42	12.82	13.25	13.71	14.74
HLRS 4030-005-160			16				60	6	10,600	16.55	17.08	17.66	18.27	19.63
HLRS 4030-005-200			20				60	6	10,600	20.67	21.34	22.06	22.82	24.53
HLRS 4030-01-040		R0.1	4				55	6	7,100	4.17	4.30	4.44	4.60	4.94
HLRS 4030-01-060			6				55	6	7,100	6.23	6.43	6.65	6.87	7.38
HLRS 4030-01-080			8				55	6	7,100	8.29	8.56	8.85	9.15	9.83
HLRS 4030-01-100			10				55	6	7,100	10.36	10.69	11.05	11.43	12.28
HLRS 4030-01-120			12				55	6	8,600	12.42	12.82	13.25	13.71	14.73
HLRS 4030-01-160			16				60	6	10,600	16.54	17.08	17.65	18.26	19.62
HLRS 4030-01-180			18				60	6	10,600	18.61	19.21	19.85	20.54	22.07
HLRS 4030-01-200			20				60	6	10,600	20.67	21.34	22.05	22.82	24.52
HLRS 4030-01-240			24				70	6	10,600	24.80	25.60	26.45	27.37	No interference
HLRS 4030-01-260			26				70	6	10,600	26.86	27.73	28.66	29.65	No interference
HLRS 4030-01-300			30				70	6	12,000	30.98	31.99	33.06	34.20	No interference
HLRS 4030-02-040		R0.2	4				55	6	7,100	4.17	4.30	4.43	4.58	4.92
HLRS 4030-02-060			6				55	6	7,100	6.23	6.43	6.64	6.86	7.36
HLRS 4030-02-080			8				55	6	7,100	8.29	8.55	8.84	9.14	9.81
HLRS 4030-02-100			10				55	6	7,100	10.35	10.68	11.04	11.42	12.26
HLRS 4030-02-120			12				55	6	8,600	12.42	12.81	13.24	13.69	14.70
HLRS 4030-02-160			16				60	6	10,600	16.54	17.07	17.64	18.25	19.60
HLRS 4030-02-180			18				60	6	10,600	18.60	19.20	19.84	20.53	22.05
HLRS 4030-02-200			20				60	6	10,600	20.67	21.33	22.04	22.80	24.49
HLRS 4030-02-240			24				70	6	10,600	24.79	25.59	26.44	27.36	No interference
HLRS 4030-02-260			26				70	6	10,600	26.86	27.72	28.65	29.63	No interference
HLRS 4030-02-300			30				70	6	12,000	30.98	31.98	33.05	34.19	No interference
HLRS 4030-02-360			36				80	6	12,000	37.17	38.37	39.65	41.02	No interference

* Additional model

HARDMAX 4 Flute Long Neck Radius End Mills

Model Number	Outside Diameter ØD	Corner Radius CR	Effective Length ℓ_1	Length of Cut ℓ	Neck Diameter Ød ₁	Shank Taper Angle Eta	Overall Length L	Shank Diameter Ød	Price ¥	Effective Length by Inclined Angles				
										30°	1°	1°30'	2°	3°
HLRS 4030-03-040	3	R0.3	4	2.4	2.92	16°	55	6	7,100	4.16	4.29	4.42	4.57	4.89
HLRS 4030-03-060			6				55	6	7,100	6.22	6.42	6.63	6.85	7.34
HLRS 4030-03-080			8				55	6	7,100	8.29	8.55	8.83	9.12	9.79
HLRS 4030-03-100			10				55	6	7,100	10.35	10.68	11.03	11.40	12.24
HLRS 4030-03-120			12				55	6	8,600	12.41	12.81	13.23	13.68	14.68
HLRS 4030-03-140			14				55	6	8,600	14.48	14.94	15.43	15.96	17.13
HLRS 4030-03-160			16				60	6	10,600	16.54	17.07	17.63	18.23	19.58
HLRS 4030-03-200			20				60	6	10,600	20.66	21.33	22.03	22.79	24.47
HLRS 4030-03-240			24				70	6	10,600	24.79	25.59	26.43	27.34	No Interference
HLRS 4030-03-260			26				70	6	10,600	26.85	27.71	28.64	29.62	No Interference
HLRS 4030-03-300			30				70	6	12,000	30.98	31.97	33.04	34.18	No Interference
HLRS 4030-03-360			36				80	6	12,000	37.17	38.36	39.64	41.01	No Interference
HLRS 4030-05-040		R0.5	4				55	6	7,100	4.15	4.27	4.40	4.54	4.85
HLRS 4030-05-060			6				55	6	7,100	6.22	6.40	6.60	6.82	7.30
HLRS 4030-05-080			8				55	6	7,100	8.28	8.53	8.80	9.10	9.74
HLRS 4030-05-100			10				55	6	7,100	10.34	10.66	11.01	11.37	12.19
HLRS 4030-05-120			12				55	6	8,600	12.40	12.79	13.21	13.65	14.64
HLRS 4030-05-160			16				60	6	10,600	16.53	17.05	17.61	18.20	19.53
HLRS 4030-05-200			20				60	6	10,600	20.66	21.31	22.01	22.76	24.43
HLRS 4030-05-240			24				70	6	10,600	24.78	25.57	26.41	27.31	29.32
HLRS 4030-05-260			26				70	6	10,600	26.84	27.70	28.61	29.59	No Interference
HLRS 4030-05-300			30				70	6	12,000	30.97	31.96	33.02	34.15	No Interference
HLRS 4030-05-360			36				80	6	12,000	37.16	38.35	39.62	40.98	No Interference
HLRS 4030-10-060		R1	6				55	6	7,100	6.20	6.37	6.55	6.75	7.18
HLRS 4030-10-080			8				55	6	7,100	8.26	8.50	8.75	9.03	9.63
HLRS 4030-10-100			10				55	6	7,100	10.32	10.63	10.95	11.30	12.08
HLRS 4030-10-120			12				55	6	8,600	12.39	12.76	13.16	13.58	14.53
HLRS 4030-10-160			16				60	6	10,600	16.51	17.02	17.56	18.13	19.42
HLRS 4030-10-200			20				60	6	10,600	20.64	21.28	21.96	22.69	24.32
HLRS 4030-10-240			24				70	6	10,600	24.76	25.54	26.36	27.24	29.21
HLRS 4030-10-260			26				70	6	10,600	26.83	27.67	28.56	29.52	No Interference
HLRS 4030-10-300			30				70	6	12,000	30.95	31.93	32.96	34.08	No Interference
HLRS 4030-10-360			36				80	6	12,000	37.14	38.31	39.57	40.91	No Interference
HLRS 4040-005-080	4	R0.05	8	3.2	3.82	16°	65	6	10,600	8.48	8.75	9.04	9.36	10.06
HLRS 4040-005-120			12				65	6	10,600	12.60	13.01	13.45	13.91	14.95
HLRS 4040-005-160			16				65	6	10,600	16.73	17.27	17.85	18.47	No Interference
HLRS 4040-005-200			20				70	6	11,800	20.85	21.53	22.25	23.02	No Interference
HLRS 4040-005-240			24				70	6	11,800	24.98	25.79	26.65	27.58	No Interference
HLRS 4040-005-320			32				80	6	11,800	33.23	34.31	35.46	No Interference	No Interference
HLRS 4040-01-080		R0.1	8				65	6	10,600	8.47	8.75	9.04	9.35	10.05
HLRS 4040-01-120			12				65	6	10,600	12.60	13.01	13.44	13.91	14.94
HLRS 4040-01-160			16				65	6	10,600	16.72	17.27	17.84	18.46	No Interference
HLRS 4040-01-200			20				70	6	11,800	20.85	21.52	22.24	23.01	No Interference
HLRS 4040-01-240			24				70	6	11,800	24.98	25.78	26.65	27.57	No Interference
HLRS 4040-01-320			32				80	6	11,800	33.23	34.30	35.45	No Interference	No Interference
HLRS 4040-01-480			48				100	6	18,800	49.73	51.34	No Interference	No Interference	No Interference

* Additional model

HARDMAX 4 Flute Long Neck Radius End Mills

Model Number	Outside Diameter ØD	Corner Radius CR	Effective Length ℓ_1	Length of Cut ℓ	Neck Diameter Ød ₁	Shank Taper Angle Eta	Overall Length L	Shank Diameter Ød	Price ¥	Effective Length by Inclined Angles				
										30°	1°	1°30'	2°	3°
HLRS 4040-02-080	4	R0.2	8	3.2	3.82	16°	65	6	10,600	8.47	8.74	9.03	9.34	10.02
HLRS 4040-02-120			12				65	6	10,600	12.60	13.00	13.43	13.89	14.92
HLRS 4040-02-160			16				65	6	10,600	16.72	17.26	17.83	18.45	No Interference
HLRS 4040-02-200			20				70	6	11,800	20.85	21.52	22.23	23.00	No Interference
HLRS 4040-02-240			24				70	6	11,800	24.97	25.78	26.64	27.56	No Interference
HLRS 4040-02-320			32				80	6	11,800	33.22	34.30	35.44	No Interference	No Interference
HLRS 4040-02-480			48				100	6	18,800	49.73	51.33	No Interference	No Interference	No Interference
HLRS 4040-03-080		R0.3	8				65	6	10,600	8.47	8.73	9.02	9.32	10.00
HLRS 4040-03-120			12				65	6	10,600	12.59	12.99	13.42	13.88	14.90
HLRS 4040-03-140			14				65	6	10,600	14.66	15.12	15.62	16.16	17.34
HLRS 4040-03-160			16				65	6	10,600	16.72	17.25	17.82	18.43	No Interference
HLRS 4040-03-200			20				70	6	11,800	20.84	21.51	22.22	22.99	No Interference
HLRS 4040-03-240			24				70	6	11,800	24.97	25.77	26.63	27.54	No Interference
HLRS 4040-03-320			32				80	6	11,800	33.22	34.29	35.43	No Interference	No Interference
HLRS 4040-03-480			48				100	6	18,800	49.72	51.33	No Interference	No Interference	No Interference
HLRS 4040-05-080		R0.5	8				65	6	10,600	8.46	8.72	9.00	9.29	9.96
HLRS 4040-05-120			12				65	6	10,600	12.58	12.98	13.40	13.85	14.85
HLRS 4040-05-160			16				65	6	10,600	16.71	17.24	17.80	18.40	19.75
HLRS 4040-05-200			20				70	6	11,800	20.84	21.50	22.20	22.96	No Interference
HLRS 4040-05-240			24				70	6	11,800	24.96	25.76	26.60	27.51	No Interference
HLRS 4040-05-320			32				80	6	11,800	33.21	34.27	35.41	No Interference	No Interference
HLRS 4040-05-400			40				100	6	17,500	41.46	42.79	No Interference	No Interference	No Interference
HLRS 4040-05-480			48				100	6	18,800	49.71	51.31	No Interference	No Interference	No Interference
HLRS 4040-10-080		R1	8				65	6	10,600	8.44	8.69	8.95	9.22	9.84
HLRS 4040-10-120			12				65	6	10,600	12.57	12.94	13.35	13.78	14.74
HLRS 4040-10-160			16				65	6	10,600	16.69	17.20	17.75	18.33	19.63
HLRS 4040-10-200			20				70	6	11,800	20.82	21.46	22.15	22.89	No Interference
HLRS 4040-10-240			24				70	6	11,800	24.94	25.72	26.55	27.44	No Interference
HLRS 4040-10-320			32				80	6	11,800	33.20	34.24	35.36	No Interference	No Interference
HLRS 4040-10-480			48				100	6	18,800	49.70	51.28	No Interference	No Interference	No Interference
HLRS 4050-005-160	5	R0.05	16	4	4.82	16°	65	6	14,300	16.73	17.27	17.85	No Interference	No Interference
HLRS 4050-005-200			20				70	6	14,000	20.85	21.53	No Interference	No Interference	No Interference
HLRS 4050-005-400			40				100	6	17,500	41.48	No Interference	No Interference	No Interference	No Interference
HLRS 4050-01-160		R0.1	16				65	6	14,300	16.72	17.27	17.84	No Interference	No Interference
HLRS 4050-01-200			20				70	6	14,000	20.85	21.52	No Interference	No Interference	No Interference
HLRS 4050-01-400			40				100	6	17,500	41.48	No Interference	No Interference	No Interference	No Interference
HLRS 4050-02-160		R0.2	16				65	6	14,000	16.72	17.26	17.83	No Interference	No Interference
HLRS 4050-02-200			20				70	6	14,000	20.85	21.52	No Interference	No Interference	No Interference
HLRS 4050-02-400			40				100	6	17,500	41.47	No Interference	No Interference	No Interference	No Interference
HLRS 4050-03-160		R0.3	16				65	6	14,000	16.72	17.25	17.82	No Interference	No Interference
HLRS 4050-03-200			20				70	6	14,000	20.84	21.51	No Interference	No Interference	No Interference
HLRS 4050-03-400			40				100	6	17,500	41.47	No Interference	No Interference	No Interference	No Interference
HLRS 4050-05-160		R0.5	16				65	6	14,000	16.71	17.24	17.80	No Interference	No Interference
HLRS 4050-05-200			20				70	6	14,000	20.83	21.50	No Interference	No Interference	No Interference
HLRS 4050-05-400			40				100	6	17,500	41.46	No Interference	No Interference	No Interference	No Interference
HLRS 4050-10-160		R1	16				65	6	14,000	16.69	17.20	17.74	No Interference	No Interference
HLRS 4050-10-200			20				70	6	14,000	20.81	21.46	No Interference	No Interference	No Interference
HLRS 4050-10-400			40				100	6	17,500	41.44	No Interference	No Interference	No Interference	No Interference

* Additional model

HARDMAX 4 Flute Long Neck Radius End Mills

* Additional model

HLRS Milling Conditions for HLRS (4 Flutes)

WORK MATERIAL			PREHARDENED STEELS / HARDENED STEELS NAK / SKD (30~45HRC)				HARDENED STEELS SKD / SKT (45~55HRC)				HARDENED STEELS SKD / SKH (55~65HRC)			
Model Number	Outside Diameter (mm)	Effective Length (mm)	Spindle Speed (min ⁻¹)	Feed Rate (mm/min)	a _p Axial Depth (mm)	a _e Radial Depth (mm)	Spindle Speed (min ⁻¹)	Feed Rate (mm/min)	a _p Axial Depth (mm)	a _e Radial Depth (mm)	Spindle Speed (min ⁻¹)	Feed Rate (mm/min)	a _p Axial Depth (mm)	a _e Radial Depth (mm)
4002	0.2	0.3	27,000	620	0.005	0.05	27,000	500	0.003	0.04	27,000	390	0.003	0.01
		0.5	24,000	500	0.005	0.05	24,000	410	0.003	0.04	24,000	250	0.003	0.01
		1	21,000	380	0.004	0.05	21,000	320	0.002	0.04	21,000	110	0.002	0.009
		1.5	18,000	260	0.004	0.05	18,000	240	0.002	0.03	18,000	80	0.002	0.007
		2	15,000	140	0.004	0.05	15,000	160	0.002	0.03	15,000	60	0.002	0.007
4003	0.3	0.3	24,500	660	0.01	0.075	23,700	530	0.006	0.07	20,000	330	0.003	0.028
		0.5	23,300	600	0.008	0.075	22,500	490	0.006	0.07	18,500	270	0.003	0.028
		1	20,400	460	0.006	0.075	19,500	400	0.005	0.07	17,000	180	0.003	0.028
		1.5	17,500	320	0.004	0.05	16,500	310	0.005	0.07	16,000	140	0.003	0.025
		2	14,600	180	0.004	0.05	13,500	220	0.005	0.07	14,500	100	0.003	0.025
4004	0.4	0.5	25,400	870	0.015	0.12	22,500	750	0.011	0.1	16,200	360	0.003	0.05
		1	22,600	700	0.012	0.12	20,500	600	0.009	0.1	15,000	250	0.003	0.045
		1.5	21,200	610	0.01	0.12	19,500	520	0.008	0.095	14,400	150	0.003	0.043
		2	19,800	530	0.009	0.12	18,500	450	0.007	0.09	13,800	100	0.003	0.04
4005	0.5	1	22,000	800	0.017	0.155	20,000	670	0.012	0.125	13,500	300	0.003	0.065
		2	19,300	610	0.013	0.155	18,000	530	0.01	0.12	12,300	170	0.003	0.06
4006	0.6	2	21,300	900	0.018	0.17	19,000	750	0.015	0.145	11,500	240	0.003	0.083
		4	18,600	690	0.013	0.17	15,500	520	0.013	0.14	10,300	50	0.003	0.07
4008	0.8	2	20,000	1,100	0.025	0.2	18,500	950	0.02	0.2	10,000	280	0.005	0.12
		3	18,800	950	0.021	0.2	16,500	830	0.018	0.2	9,200	200	0.005	0.116
		4	17,500	840	0.018	0.2	15,000	730	0.016	0.2	8,800	120	0.004	0.112
		6	14,600	700	0.015	0.2	12,500	600	0.015	0.2	8,500	60	0.004	0.108
		8	13,100	450	0.008	0.13	11,150	425	0.008	0.125	7,500	50	0.003	0.057
4010	1	2	17,600	1,470	0.056	0.27	15,300	1,200	0.038	0.27	10,900	710	0.03	0.27
		3	15,500	1,390	0.048	0.27	13,200	1,150	0.037	0.27	9,400	680	0.027	0.27
		4	13,800	1,310	0.039	0.27	12,000	1,070	0.031	0.243	8,500	640	0.015	0.243
		5	12,500	1,150	0.03	0.24	11,000	960	0.027	0.232	7,800	570	0.013	0.144
		6	11,300	1,040	0.021	0.216	9,800	860	0.016	0.209	7,000	510	0.01	0.108
		8	9,800	780	0.02	0.189	8,500	720	0.012	0.16	6,100	420	0.008	0.094
		10	8,800	510	0.011	0.126	7,600	510	0.009	0.1	5,400	350	0.006	0.05
		12	8,100	320	0.008	0.1	7,000	400	0.006	0.05	5,000	300	0.006	0.03
		16	7,000	150	0.005	0.06	6,100	330	0.006	0.05	4,300	250	0.005	0.01
4012	1.2	4	13,200	1,360	0.032	0.45	11,900	1,100	0.024	0.3	9,200	1,300	0.02	0.2
		6	11,200	1,160	0.028	0.36	9,600	980	0.022	0.252	7,400	1,200	0.011	0.095
		10	9,000	800	0.017	0.18	7,300	600	0.009	0.15	6,300	800	0.006	0.05
4015	1.5	3	16,400	1,520	0.063	0.569	14,800	1,330	0.052	0.54	11,200	780	0.035	0.315
		4	13,200	1,360	0.054	0.54	13,200	1,280	0.042	0.495	10,100	700	0.033	0.292
		6	11,600	1,280	0.041	0.486	10,600	1,210	0.038	0.445	8,100	460	0.025	0.202
		8	10,200	1,080	0.037	0.378	9,300	1,020	0.031	0.346	7,100	390	0.015	0.157
		10	9,300	930	0.032	0.345	8,500	870	0.029	0.316	6,600	340	0.011	0.172
		12	8,500	830	0.029	0.324	7,800	780	0.026	0.297	5,900	300	0.01	0.162
		16	7,400	670	0.018	0.216	6,800	600	0.014	0.198	5,100	230	0.005	0.108
		18	6,000	550	0.015	0.2	5,800	420	0.01	0.15	4,200	150	0.005	0.08

HLRS Milling Conditions for HLRS (4 Flutes)

WORK MATERIAL			PREHARDENED STEELS / HARDENED STEELS NAK / SKD (30~45HRC)				HARDENED STEELS SKD / SKT (45~55HRC)				HARDENED STEELS SKD / SKH (55~65HRC)			
Model Number	Outside Diameter (mm)	Effective Length (mm)	Spindle Speed (min ⁻¹)	Feed Rate (mm/min)	a _p Axial Depth (mm)	a _e Radial Depth (mm)	Spindle Speed (min ⁻¹)	Feed Rate (mm/min)	a _p Axial Depth (mm)	a _e Radial Depth (mm)	Spindle Speed (min ⁻¹)	Feed Rate (mm/min)	a _p Axial Depth (mm)	a _e Radial Depth (mm)
4018	1.8	8	10,700	1,120	0.047	0.495	9,800	1,060	0.043	0.497	7,700	500	0.02	0.222
		10	9,600	1,010	0.04	0.436	8,900	950	0.038	0.421	7,100	390	0.015	0.203
		12	8,100	850	0.035	0.303	7,500	740	0.032	0.306	5,900	290	0.013	0.159
		14	7,400	770	0.027	0.24	6,900	660	0.024	0.24	5,400	270	0.008	0.13
		16	7,200	730	0.021	0.207	6,700	630	0.019	0.198	5,200	260	0.006	0.113
4020	2	4	15,300	1,570	0.069	0.72	14,300	1,460	0.065	0.81	11,500	860	0.031	0.36
		6	12,800	1,280	0.064	0.648	12,000	1,200	0.06	0.729	9,700	700	0.028	0.324
		8	11,200	1,160	0.058	0.612	10,400	1,100	0.055	0.648	8,400	600	0.026	0.288
		10	10,000	1,090	0.049	0.526	9,300	1,020	0.047	0.526	7,600	450	0.019	0.234
		12	9,100	1,030	0.046	0.405	8,500	960	0.044	0.405	6,900	420	0.018	0.18
		16	7,800	860	0.042	0.283	7,300	700	0.039	0.315	5,900	270	0.016	0.157
		20	7,000	800	0.025	0.196	6,600	650	0.024	0.198	5,300	290	0.007	0.118
		24	6,500	740	0.02	0.17	5,600	500	0.019	0.14	4,700	270	0.007	0.1
		26	6,000	680	0.013	0.15	5,000	440	0.012	0.12	4,000	220	0.005	0.08
4025	2.5	30	5,500	620	0.01	0.1	4,400	390	0.009	0.09	3,400	180	0.005	0.08
		6	13,000	1,600	0.078	0.7	12,000	1,500	0.074	0.7	9,900	830	0.05	0.476
		8	11,300	1,430	0.075	0.62	10,500	1,240	0.072	0.62	9,100	650	0.05	0.42
		10	10,500	1,400	0.067	0.54	10,000	1,150	0.067	0.54	8,400	510	0.048	0.324
		16	8,900	1,400	0.059	0.36	8,500	790	0.049	0.3	7,200	350	0.03	0.15
		20	7,800	1,200	0.048	0.27	7,500	670	0.031	0.225	6,300	300	0.022	0.09
4030	3	30	6,300	600	0.011	0.18	6,000	500	0.014	0.18	5,000	220	0.01	0.054
		4	15,000	3,070	0.128	0.72	14,000	2,640	0.08	0.72	11,500	980	0.052	0.576
		6	14,000	2,890	0.12	0.72	13,300	2,500	0.075	0.72	10,800	900	0.05	0.576
		8	12,500	2,530	0.105	0.7	11,800	2,200	0.07	0.7	9,900	810	0.047	0.56
		10	11,300	2,160	0.096	0.7	10,500	2,090	0.06	0.7	9,000	730	0.045	0.56
		12	10,500	2,020	0.084	0.67	10,000	1,950	0.052	0.67	8,100	660	0.037	0.502
		14	9,700	1,800	0.072	0.65	9,300	1,700	0.044	0.65	7,500	600	0.032	0.43
		16	9,200	1,680	0.064	0.63	8,800	1,600	0.04	0.63	7,100	570	0.027	0.378
		18	8,600	1,610	0.061	0.605	8,300	1,540	0.038	0.605	6,700	560	0.023	0.348
		20	8,400	1,540	0.058	0.58	7,900	1,490	0.036	0.58	6,300	550	0.022	0.319
		24	7,900	1,490	0.052	0.513	7,500	1,420	0.029	0.513	6,000	510	0.017	0.258
		26	7,500	1,440	0.046	0.446	7,100	1,360	0.023	0.446	5,700	480	0.012	0.198
4040	4	30	7,000	1,260	0.04	0.38	6,500	1,230	0.015	0.38	5,400	390	0.007	0.144
		36	6,500	1,100	0.035	0.3	6,100	1,110	0.01	0.18	5,100	300	0.005	0.1
		8	10,200	1,480	0.133	1.35	8,500	1,420	0.104	1.35	7,300	810	0.091	0.945
		12	8,900	1,440	0.116	1.15	7,600	1,390	0.091	1.15	6,400	780	0.065	0.805
		14	8,500	1,400	0.1	1.08	7,100	1,350	0.078	1.08	6,000	760	0.051	0.76
		16	7,900	1,370	0.091	1	6,600	1,330	0.071	1	5,600	740	0.043	0.7
		20	6,900	1,320	0.076	0.9	5,800	1,260	0.059	0.9	4,900	720	0.032	0.63
		24	6,200	1,200	0.06	0.8	5,200	1,120	0.047	0.8	4,500	630	0.022	0.56
		32	5,500	960	0.037	0.648	4,600	920	0.029	0.648	3,900	600	0.011	0.388
		40	5,000	790	0.025	0.55	4,300	800	0.02	0.482	3,600	560	0.008	0.3
		48	4,600	700	0.012	0.45	3,900	680	0.01	0.315	3,300	500	0.005	0.1

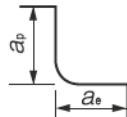
HLRS Milling Conditions for HLRS (4 Flutes)

WORK MATERIAL			PREHARDENED STEELS / HARDENED STEELS NAK / SKD (30~45HRC)				HARDENED STEELS SKD / SKT (45~55HRC)				HARDENED STEELS SKD / SKH (55~65HRC)			
Model Number	Outside Diameter (mm)	Effective Length (mm)	Spindle Speed (min ⁻¹)	Feed Rate (mm/min)	a_p Axial Depth (mm)	a_e Radial Depth (mm)	Spindle Speed (min ⁻¹)	Feed Rate (mm/min)	a_p Axial Depth (mm)	a_e Radial Depth (mm)	Spindle Speed (min ⁻¹)	Feed Rate (mm/min)	a_p Axial Depth (mm)	a_e Radial Depth (mm)
4050	5	16	7,200	1,700	0.15	1.5	5,300	1,200	0.125	1.15	4,200	820	0.063	1.03
		20	6,700	1,500	0.14	1.4	4,800	1,100	0.1	1.1	4,000	740	0.045	1
		40	4,600	880	0.068	1	3,300	700	0.06	0.68	2,800	500	0.025	0.44
4060	6	12	8,000	2,370	0.2	2.43	4,700	1,360	0.2	1.35	4,000	1,080	0.075	1.35
		16	6,700	2,020	0.19	2.394	4,000	1,150	0.19	1.33	3,400	900	0.073	1.33
		18	6,300	1,890	0.185	2.376	3,750	1,060	0.185	1.32	3,200	830	0.071	1.32
		20	5,800	1,730	0.18	2.358	3,500	1,000	0.18	1.31	3,000	760	0.07	1.31
		24	5,200	1,540	0.17	2.322	3,100	860	0.17	1.29	2,700	680	0.068	1.29
		30	4,500	1,290	0.158	2.268	2,600	740	0.158	1.26	2,200	580	0.066	1.26
		40	3,000	800	0.1	1.35	1,700	480	0.1	0.75	1,400	360	0.04	0.55
		48	2,000	510	0.05	0.9	1,200	330	0.04	0.5	1,000	240	0.02	0.3

Side Milling

a_p : Axial Depth (mm)

a_e : Radial Depth (mm)



Note:

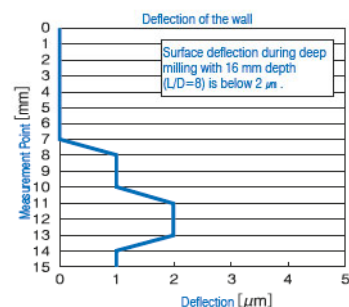
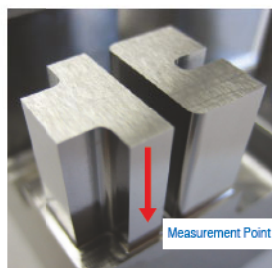
- Decrease both spindle speed and feed rate proportionally when the milling parameters exceed the machine's maximum spindle speed
- Every coolant offers stable milling.
- Recommend oil coolant for Stainless Steels and Heat Resistant Alloys.
- Recommend wet coolant for Copper.

Milling Example for Slotting

HLRS4000 Ø2 × CR0.3 × **Effective Length 16** (L / D = 8)

DAC10 (48HRC)

Tool	HLRS 4020-03-160
Spindle Speed	7,300 min ⁻¹
Feed Rate	1,260 mm/min
Axial Depth a_p	0.02 mm
Radial Depth a_e	0.015 mm
Cycle Time	112 min
Coolant	Oil Mist
Work Material Size	20 × 15 mm



HARDMAX 2 Flute Long Neck Radius End Mills

Size $\varnothing 0.2 \sim \varnothing 6$



HLRS2000/HLRS2000E

Super
MG

HARD
MAX

30°

R

R
±0.005

Shank Dia
0/-0.005

Material Applications (☆ Highly Recommended ◎ Recommended ○ Suggested)

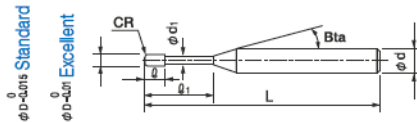
Work Material															
CARBON STEELS S45C S55C	ALLOY STEELS SK / SCM SUS	PREHARDENED STEELS NAK HPM	HARDENED STEELS			CAST IRON	ALUMINUM ALLOYS	GRAPHITE	COPPER	PLASTICS	GLASS FILLED PLASTICS	TITANIUM ALLOYS	HEAT RESISTANT ALLOYS	CEMENTED CARBIDE	HARD BRITTLE (NON-METALLIC) MATERIALS
			~ 55HRC	~ 60HRC	~ 70HRC										
○	○	◎	◎	◎	○ ~ 65HRC	○			◎			○	○		

Features

Long Neck Radius design for high efficiency and high quality milling.

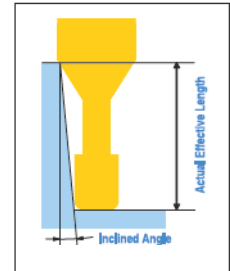
Recommended for various applications from Copper and Raw Materials to Hard Materials.

Both dry and wet coolant offer stable and long tool life.



The shank taper angle shown is not an exact value and to avoid contact with the work piece, we recommend the user controls the precise value of this angle. Shank taper angle should not make contact with the work piece.

	Diameter Tolerance	Corner Radius Tolerance
Standard Tolerance Type	0/-0.015	Nominal Radius ± 0.005
Excellent Tolerance Type	0/-0.01	Nominal Radius ± 0.005



Total 353 models

Unit (mm)

Model Number	Excellent	Outside Diameter $\varnothing D$	Corner Radius CR	Effective Length ℓ_1	Length of Cut ℓ	Neck Diameter $\varnothing d_1$	Shank Taper Angle Bta	Overall Length L	Shank Diameter $\varnothing d$	Price ¥	Effective Length by Inclined Angles				
											30°	1°	1°30'	2°	3°
HLRS 2002-005-005E	○	0.2	R0.05	0.5	0.2	0.17	16°	50	4	12,320	0.67	0.71	0.75	0.78	0.85
HLRS 2002-005-010E	○			1				50	4	12,320	1.20	1.26	1.31	1.36	1.45
HLRS 2002-005-015E	○			1.5				50	4	12,320	1.72	1.80	1.87	1.92	2.03
HLRS 2002-005-020E	○			2				50	4	12,320	2.25	2.34	2.41	2.48	2.59
HLRS 2003-005-010E	○	0.3	R0.05	1	0.3	0.27	16°	50	4	11,870	1.24	1.31	1.38	1.44	1.55
HLRS 2003-005-015E	○			1.5				50	4	11,870	1.72	1.83	1.91	1.99	2.12
HLRS 2003-005-020E	○			2				50	4	11,870	2.26	2.37	2.47	2.55	2.70
HLRS 2003-005-025E	○			2.5				50	4	13,200	2.78	2.91	3.02	3.11	3.27
HLRS 2003-005-030E	○	0.4	R0.05	3	0.4	0.38	16°	50	4	13,200	3.31	3.45	3.57	3.66	3.83
HLRS 2004-005-010E	○			1				50	4	7,910	1.31	1.40	1.49	1.57	1.72
HLRS 2004-005-015E	○			1.5				50	4	7,910	1.79	1.92	2.03	2.13	2.31
HLRS 2004-005-020E	○			2				50	4	7,910	2.33	2.48	2.60	2.71	2.90
HLRS 2004-005-030E	○			3				50	4	7,910	3.40	3.58	3.72	3.85	4.07
HLRS 2004-005-040E	○			4				50	4	7,910	4.45	4.66	4.82	4.97	5.21
HLRS 2004-01-010			R0.1	1				50	4	7,910	1.28	1.38	1.46	1.55	1.69
HLRS 2004-01-015				1.5				50	4	7,910	1.76	1.90	2.01	2.11	2.28
HLRS 2004-01-020				2				50	4	7,910	2.30	2.46	2.58	2.69	2.89
HLRS 2004-01-030				3				50	4	7,910	3.38	3.56	3.71	3.83	4.06
HLRS 2004-01-040				4				50	4	7,910	4.44	4.64	4.81	4.95	5.20

HARDMAX 2 Flute Long Neck Radius End Mills

Model Number	Excellent	Outside Diameter ØD	Corner Radius CR	Effective Length ℓ_1	Length of Cut ℓ	Neck Diameter Ød ₁	Shank Taper Angle Bta	Overall Length L	Shank Diameter Ød	Price ¥	Effective Length by Inclined Angles				
											30°	1°	1°30'	2°	3°
HLRS 2005-005-010		0.5	R0.05	1	0.5	0.48	16°	50	4	6,440	1.34	1.46	1.57	1.67	1.86
HLRS 2005-005-020				2				50	4	6,440	2.37	2.55	2.71	2.84	3.08
HLRS 2005-005-030				3				50	4	6,440	3.45	3.67	3.85	4.00	4.27
HLRS 2005-005-040				4				50	4	6,440	4.52	4.77	4.97	5.14	5.44
HLRS 2005-005-050				5				50	4	6,440	5.58	5.85	6.07	6.26	6.58
HLRS 2005-01-010			R0.1	1				50	4	6,440	1.34	1.45	1.56	1.66	1.85
HLRS 2005-01-020				2				50	4	6,440	2.37	2.55	2.70	2.83	3.07
HLRS 2005-01-030				3				50	4	6,440	3.45	3.67	3.84	4.00	4.26
HLRS 2005-01-040				4				50	4	6,440	4.52	4.76	4.96	5.13	5.43
HLRS 2005-01-050				5				50	4	6,440	5.58	5.85	6.07	6.25	6.57
HLRS 2005-01-060				6				50	4	6,440	6.63	6.93	7.16	7.36	7.70
HLRS 2008-005-020		0.6	R0.05	2	0.6	0.58	16°	50	4	6,440	2.38	2.61	2.79	2.95	3.22
HLRS 2008-005-030				3				50	4	6,440	3.48	3.74	3.95	4.13	4.30
HLRS 2008-005-040				4				50	4	6,440	4.56	4.85	5.08	5.28	5.67
HLRS 2008-005-060				6				50	4	6,440	6.68	7.03	7.30	7.55	8.12
HLRS 2008-005-080				8				50	4	6,440	8.79	9.18	9.50	9.83	10.56
HLRS 2008-01-020			R0.1	2				50	4	6,440	2.37	2.60	2.78	2.93	3.20
HLRS 2008-01-020E	○			2				50	4	7,080	2.37	2.60	2.78	2.93	3.20
HLRS 2008-01-030				3				50	4	6,440	3.47	3.73	3.94	4.11	4.28
HLRS 2008-01-030E	○			3				50	4	7,080	3.47	3.73	3.94	4.11	4.28
HLRS 2008-01-040				4				50	4	6,440	4.55	4.84	5.07	5.26	5.65
HLRS 2008-01-040E	○			4				50	4	7,080	4.55	4.84	5.07	5.26	5.65
HLRS 2008-01-060			R0.2	6				50	4	6,440	6.68	7.03	7.30	7.54	8.10
HLRS 2008-01-080				8				50	4	6,440	8.79	9.18	9.50	9.82	10.55
HLRS 2008-02-020				2				50	4	6,440	2.34	2.56	2.74	2.90	3.18
HLRS 2008-02-030				3				50	4	6,440	3.44	3.70	3.91	4.09	4.41
HLRS 2008-02-040				4				50	4	6,440	4.53	4.82	5.05	5.23	5.61
HLRS 2008-02-060				6				50	4	6,440	6.66	7.01	7.28	7.51	8.06
HLRS 2008-02-080				8				50	4	6,440	8.79	9.17	9.48	9.81	10.53
HLRS 2007-01-040		0.7	R0.1	4	0.7	0.68	16°	50	4	6,780	4.55	4.84	5.07	5.26	5.65
HLRS 2007-01-060				6				50	4	6,780	6.68	7.03	7.30	7.54	8.10
HLRS 2007-02-040			R0.2	4				50	4	6,780	4.53	4.82	5.05	5.23	5.61
HLRS 2007-02-060				6				50	4	6,780	6.66	7.01	7.28	7.51	8.06
HLRS 2008-005-040		0.8	R0.05	4	0.8	0.78	16°	50	4	7,340	4.56	4.85	5.08	5.28	5.67
HLRS 2008-005-060				6				50	4	7,340	6.68	7.03	7.30	7.55	8.12
HLRS 2008-005-080				8				50	4	7,340	8.79	9.18	9.50	9.83	10.56
HLRS 2008-01-040			R0.1	4				50	4	7,340	4.55	4.84	5.07	5.26	5.65
HLRS 2008-01-060				6				50	4	7,340	6.68	7.03	7.30	7.54	8.10
HLRS 2008-01-080				8				50	4	7,340	8.79	9.18	9.50	9.82	10.55
HLRS 2008-02-040			R0.2	4				50	4	7,340	4.53	4.82	5.05	5.23	5.61
HLRS 2008-02-060				6				50	4	7,340	6.66	7.01	7.28	7.51	8.06
HLRS 2008-02-080				8				50	4	7,340	8.79	9.17	9.48	9.81	10.53
HLRS 2010-005-020		1	R0.05	2	1	0.95	16°	50	4	6,240	2.51	2.86	2.70	3.01	3.28
HLRS 2010-005-030				3				50	4	6,240	3.59	3.82	4.01	4.18	4.51
HLRS 2010-005-040				4				50	4	6,240	4.72	4.92	5.14	5.33	5.73
HLRS 2010-005-050				5				50	4	6,240	5.72	6.01	6.25	6.47	6.95
HLRS 2010-005-060				6				50	4	6,780	6.77	7.09	7.35	7.61	8.18
HLRS 2010-005-080				8				50	4	6,780	8.87	9.24	9.55	9.88	10.62
HLRS 2010-005-100				10				50	4	6,780	10.97	11.37	11.75	12.16	13.07
HLRS 2010-005-120				12				55	4	6,780	13.05	13.50	13.96	14.44	15.52
HLRS 2010-005-160				16				60	4	8,990	17.20	17.76	18.36	18.99	20.41
HLRS 2010-005-200				20				60	4	9,980	21.33	22.02	22.76	23.55	25.31

HARDMAX 2 Flute Long Neck Radius End Mills

Model Number	Excellent	Outside Diameter ØD	Corner Radius CR	Effective Length ℓ_1	Length of Cut ℓ	Neck Diameter Ød ₁	Shank Taper Angle Bta	Overall Length L	Shank Diameter Ød	Price ¥	Effective Length by Inclined Angles				
											30°	1°	1°30'	2°	3°
HLRS 2010-01-020		1	R0.1	2	1	0.95	16°	50	4	6,240	2.53	2.71	2.88	3.01	3.27
HLRS 2010-01-020E	○			2				50	4	6,860	2.53	2.71	2.88	3.01	3.27
HLRS 2010-01-030				3				50	4	6,240	3.58	3.81	4.00	4.18	4.49
HLRS 2010-01-040				4				50	4	6,240	4.67	4.93	5.14	5.33	5.72
HLRS 2010-01-040E	○			4				50	4	6,860	4.67	4.93	5.14	5.33	5.72
HLRS 2010-01-050				5				50	4	6,240	5.71	6.00	6.24	6.46	6.94
HLRS 2010-01-080				6				50	4	6,780	6.78	7.10	7.36	7.60	8.17
HLRS 2010-01-080E	○			6				50	4	7,460	6.78	7.10	7.36	7.60	8.17
HLRS 2010-01-080				8				50	4	6,780	8.88	9.24	9.56	9.88	10.61
HLRS 2010-01-100				10				50	4	6,780	10.97	11.37	11.76	12.16	13.06
HLRS 2010-01-120				12				55	4	6,780	13.05	13.50	13.96	14.44	15.51
HLRS 2010-01-160				16				60	4	8,990	17.20	17.76	18.36	18.99	20.40
HLRS 2010-01-200				20				60	4	9,980	21.33	22.02	22.76	23.54	25.30
HLRS 2010-02-020			R0.2	2				50	4	6,240	2.51	2.69	2.86	2.98	3.23
HLRS 2010-02-020E	○			2				50	4	6,860	2.51	2.69	2.86	2.98	3.23
HLRS 2010-02-030				3				50	4	6,240	3.58	3.80	3.99	4.16	4.47
HLRS 2010-02-040				4				50	4	6,240	4.65	4.91	5.12	5.30	5.68
HLRS 2010-02-040E	○			4				50	4	6,860	4.65	4.91	5.12	5.30	5.68
HLRS 2010-02-050				5				50	4	6,240	5.71	6.00	6.23	6.45	6.92
HLRS 2010-02-080				6				50	4	6,780	6.76	7.08	7.34	7.57	8.13
HLRS 2010-02-080E	○			6				50	4	7,460	6.76	7.08	7.34	7.57	8.13
HLRS 2010-02-080				8				50	4	6,780	8.86	9.22	9.54	9.85	10.57
HLRS 2010-02-100				10				50	4	6,780	10.95	11.35	11.74	12.13	13.02
HLRS 2010-02-120				12				55	4	6,780	13.03	13.48	13.94	14.41	15.47
HLRS 2010-02-160				16				60	4	8,990	17.18	17.74	18.34	18.96	20.36
HLRS 2010-02-200				20				60	4	9,980	21.31	22.00	22.74	23.51	25.26
HLRS 2010-03-020			R0.3	2				50	4	6,240	2.49	2.67	2.84	2.95	3.19
HLRS 2010-03-020E	○			2				50	4	6,860	2.49	2.67	2.84	2.95	3.19
HLRS 2010-03-030				3				50	4	6,240	3.57	3.79	3.98	4.14	4.45
HLRS 2010-03-040				4				50	4	6,240	4.63	4.89	5.10	5.27	5.64
HLRS 2010-03-040E	○			4				50	4	6,860	4.63	4.89	5.10	5.27	5.64
HLRS 2010-03-050				5				50	4	6,240	5.70	5.99	6.22	6.43	6.90
HLRS 2010-03-080				6				50	4	6,780	6.74	7.06	7.32	7.54	8.09
HLRS 2010-03-080E	○			6				50	4	7,460	6.74	7.06	7.32	7.54	8.09
HLRS 2010-03-080				8				50	4	6,780	8.84	9.20	9.52	9.82	10.53
HLRS 2010-03-100				10				50	4	6,780	10.93	11.33	11.72	12.10	12.98
HLRS 2010-03-120				12				55	4	6,780	13.01	13.46	13.92	14.38	15.43
HLRS 2010-03-160				16				60	4	8,990	17.16	17.72	18.32	18.93	20.32
HLRS 2010-03-200				20				60	4	9,980	21.29	21.98	22.72	23.48	25.22
HLRS 2012-02-080		1.2	R0.2	6	1.2	1.14	16°	50	4	7,000	6.18	6.38	6.59	6.82	7.33
HLRS 2012-02-120				12				55	4	7,000	12.37	12.77	13.19	13.65	14.67
HLRS 2012-02-200				20				60	4	10,620	20.62	21.29	22.00	22.76	24.46
HLRS 2012-03-080			R0.3	6				50	4	7,000	6.18	6.38	6.59	6.81	7.31
HLRS 2012-03-120				12				55	4	7,000	12.37	12.77	13.19	13.64	14.66
HLRS 2012-03-200				20				60	4	10,620	20.62	21.28	21.99	22.75	24.45
HLRS 2015-005-040		1.5	R0.05	4	1.5	1.45	16°	50	4	6,650	4.12	4.26	4.40	4.55	4.89
HLRS 2015-005-080				6				50	4	6,650	6.18	6.39	6.60	6.83	7.34
HLRS 2015-005-080				8				50	4	7,000	8.25	8.52	8.80	9.11	9.79
HLRS 2015-005-100				10				50	4	7,000	10.31	10.64	11.00	11.38	12.24

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Model Number	Excellent	Outside Diameter ØD	Corner Radius CR	Effective Length ℓ_1	Length of Cut ℓ	Neck Diameter Ød ₁	Shank Taper Angle Bta	Overall Length L	Shank Diameter Ød	Price ¥	Effective Length by Inclined Angles				
											30°	1°	1°30'	2°	3°
HLRS 2015-01-040		1.5	R0.1	4	1.5	1.45	16°	50	4	6,650	4.12	4.25	4.40	4.55	4.89
HLRS 2015-01-060				6				50	4	6,650	6.18	6.38	6.60	6.83	7.34
HLRS 2015-01-080				8				50	4	7,000	8.24	8.51	8.80	9.10	9.78
HLRS 2015-01-100				10				50	4	7,000	10.31	10.64	11.00	11.38	12.23
HLRS 2015-01-120				12				55	4	7,000	12.37	12.77	13.20	13.66	14.68
HLRS 2015-01-160				16				55	4	7,000	16.50	17.03	17.60	18.21	19.57
HLRS 2015-01-200				20				60	4	7,000	20.62	21.29	22.00	22.77	No Interference
HLRS 2015-02-040			R0.2	4				50	4	6,650	4.12	4.25	4.39	4.54	4.88
HLRS 2015-02-060				6				50	4	6,650	6.18	6.38	6.59	6.82	7.33
HLRS 2015-02-080				8				50	4	7,000	8.24	8.51	8.79	9.10	9.77
HLRS 2015-02-100				10				50	4	7,000	10.31	10.64	10.99	11.37	12.22
HLRS 2015-02-120				12				55	4	7,000	12.37	12.77	13.19	13.65	14.67
HLRS 2015-02-160				16				55	4	7,000	16.49	17.03	17.60	18.21	19.56
HLRS 2015-02-200				20				60	4	7,000	20.62	21.29	22.00	22.76	No Interference
HLRS 2015-03-040			R0.3	4				50	4	6,650	4.12	4.25	4.39	4.54	4.87
HLRS 2015-03-060				6				50	4	6,650	6.18	6.38	6.59	6.81	7.31
HLRS 2015-03-080				8				50	4	7,000	8.24	8.51	8.79	9.09	9.76
HLRS 2015-03-100				10				50	4	7,000	10.30	10.64	10.99	11.37	12.21
HLRS 2015-03-120				12				55	4	7,000	12.37	12.77	13.19	13.64	14.66
HLRS 2015-03-160				16				55	4	7,000	16.49	17.02	17.59	18.20	19.55
HLRS 2015-03-200				20				60	4	7,000	20.62	21.28	21.99	22.75	No Interference
HLRS 2015-05-040			R0.5	4				50	4	6,650	4.11	4.24	4.38	4.52	4.85
HLRS 2015-05-060				6				50	4	6,650	6.18	6.37	6.58	6.80	7.29
HLRS 2015-05-080				8				50	4	7,000	8.24	8.50	8.78	9.08	9.74
HLRS 2015-05-100				10				50	4	7,000	10.30	10.63	10.98	11.35	12.19
HLRS 2015-05-120				12				55	4	7,000	12.36	12.76	13.18	13.63	14.64
HLRS 2015-05-160				16				55	4	7,000	16.49	17.02	17.58	18.19	19.53
HLRS 2015-05-200				20				60	4	7,000	20.62	21.28	21.98	22.74	24.42
HLRS 2020-005-040		2	R0.05	4	2	1.92	16°	50	4	6,650	4.16	4.29	4.44	4.59	4.94
HLRS 2020-005-060				6				50	4	6,650	6.22	6.42	6.64	6.87	7.38
HLRS 2020-005-080				8				50	4	7,000	8.28	8.55	8.84	9.15	9.83
HLRS 2020-005-100				10				50	4	7,000	10.35	10.68	11.04	11.42	12.28
HLRS 2020-01-040			R0.1	4				50	4	6,650	4.16	4.29	4.43	4.59	4.93
HLRS 2020-01-040E	○			4				50	4	7,320	4.16	4.29	4.43	4.59	4.93
HLRS 2020-01-060				6				50	4	6,650	6.22	6.42	6.64	6.87	7.38
HLRS 2020-01-060E	○			6				50	4	7,320	6.22	6.42	6.64	6.87	7.38
HLRS 2020-01-080				8				50	4	7,000	8.28	8.55	8.84	9.14	9.83
HLRS 2020-01-080E	○			8				50	4	7,700	8.28	8.55	8.84	9.14	9.83
HLRS 2020-01-100				10				50	4	7,000	10.34	10.68	11.04	11.42	12.27
HLRS 2020-01-100E	○			10				50	4	7,700	10.34	10.68	11.04	11.42	12.27
HLRS 2020-01-120				12				55	4	7,000	12.41	12.81	13.24	13.70	14.72
HLRS 2020-01-120E	○			12				55	4	7,700	12.41	12.81	13.24	13.70	14.72
HLRS 2020-01-160				16				60	4	7,000	16.53	17.07	17.64	18.25	No Interference
HLRS 2020-01-200				20				60	4	7,000	20.66	21.33	22.04	22.81	No Interference
HLRS 2020-01-260				26				70	4	7,000	26.85	27.72	28.65	No Interference	No Interference
HLRS 2020-01-300				30				70	4	7,000	30.97	31.98	33.05	No Interference	No Interference

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Model Number	Excellent	Outside Diameter ØD	Corner Radius CR	Effective Length ℓ_1	Length of Cut ℓ	Neck Diameter Ød ₁	Shank Taper Angle Bta	Overall Length L	Shank Diameter Ød	Price ¥	Effective Length by Inclined Angles				
											30°	1°	1°30'	2°	3°
HLRS 2020-02-040		2	R0.2	4	2	1.92	16°	50	4	6,650	4.15	4.29	4.43	4.58	4.92
HLRS 2020-02-040E	○			4				50	4	7,320	4.15	4.29	4.43	4.58	4.92
HLRS 2020-02-060				6				50	4	6,650	6.22	6.42	6.63	6.86	7.37
HLRS 2020-02-060E	○			6				50	4	7,320	6.22	6.42	6.63	6.86	7.37
HLRS 2020-02-080				8				50	4	7,000	8.28	8.55	8.83	9.14	9.82
HLRS 2020-02-080E	○			8				50	4	7,700	8.28	8.55	8.83	9.14	9.82
HLRS 2020-02-100				10				50	4	7,000	10.34	10.68	11.03	11.41	12.26
HLRS 2020-02-100E	○			10				50	4	7,700	10.34	10.68	11.03	11.41	12.26
HLRS 2020-02-120				12				55	4	7,000	12.40	12.81	13.23	13.69	14.71
HLRS 2020-02-120E	○			12				55	4	7,700	12.40	12.81	13.23	13.69	14.71
HLRS 2020-02-160				16				60	4	7,000	16.53	17.06	17.64	18.25	No Interference
HLRS 2020-02-200				20				60	4	7,000	20.66	21.32	22.04	22.80	No Interference
HLRS 2020-02-260				26				70	4	7,000	26.84	27.71	28.64	No Interference	No Interference
HLRS 2020-02-300				30				70	4	7,000	30.97	31.97	33.04	No Interference	No Interference
HLRS 2020-03-040			R0.3	4				50	4	6,650	4.15	4.28	4.42	4.57	4.91
HLRS 2020-03-040E	○			4				50	4	7,320	4.15	4.28	4.42	4.57	4.91
HLRS 2020-03-060				6				50	4	6,650	6.21	6.41	6.63	6.85	7.36
HLRS 2020-03-060E	○			6				50	4	7,320	6.21	6.41	6.63	6.85	7.36
HLRS 2020-03-080				8				50	4	7,000	8.28	8.54	8.83	9.13	9.80
HLRS 2020-03-080E	○			8				50	4	7,700	8.28	8.54	8.83	9.13	9.80
HLRS 2020-03-100				10				50	4	7,000	10.34	10.67	11.03	11.41	12.25
HLRS 2020-03-100E	○			10				50	4	7,700	10.34	10.67	11.03	11.41	12.25
HLRS 2020-03-120				12				55	4	7,000	12.40	12.80	13.23	13.68	14.70
HLRS 2020-03-120E	○			12				55	4	7,700	12.40	12.80	13.23	13.68	14.70
HLRS 2020-03-160				16				60	4	7,000	16.53	17.06	17.63	18.24	19.59
HLRS 2020-03-200				20				60	4	7,000	20.65	21.32	22.03	22.79	No Interference
HLRS 2020-03-260				26				70	4	7,000	26.84	27.71	28.64	No Interference	No Interference
HLRS 2020-03-300				30				70	4	7,000	30.97	31.97	33.04	No Interference	No Interference
HLRS 2020-05-040			R0.5	4				50	4	6,650	4.15	4.28	4.41	4.56	4.89
HLRS 2020-05-040E	○			4				50	4	7,320	4.15	4.28	4.41	4.56	4.89
HLRS 2020-05-060				6				50	4	6,650	6.21	6.41	6.62	6.84	7.34
HLRS 2020-05-060E	○			6				50	4	7,320	6.21	6.41	6.62	6.84	7.34
HLRS 2020-05-080				8				50	4	7,000	8.27	8.54	8.82	9.12	9.78
HLRS 2020-05-080E	○			8				50	4	7,700	8.27	8.54	8.82	9.12	9.78
HLRS 2020-05-100				10				50	4	7,000	10.34	10.67	11.02	11.39	12.23
HLRS 2020-05-100E	○			10				50	4	7,700	10.34	10.67	11.02	11.39	12.23
HLRS 2020-05-120				12				55	4	7,000	12.40	12.80	13.22	13.67	14.68
HLRS 2020-05-120E	○			12				55	4	7,700	12.40	12.80	13.22	13.67	14.68
HLRS 2020-05-160				16				60	4	7,000	16.53	17.06	17.62	18.23	19.57
HLRS 2020-05-200				20				60	4	7,000	20.65	21.31	22.02	22.78	No Interference
HLRS 2020-05-260				26				70	4	7,000	26.84	27.70	28.63	No Interference	No Interference
HLRS 2020-05-300				30				70	4	7,000	30.97	31.96	33.03	No Interference	No Interference
HLRS 2025-01-100		2.5	R0.1	10	2.5	2.42	16°	50	4	7,340	10.34	10.68	11.04	11.42	12.27
HLRS 2025-01-200				20				60	4	7,570	20.66	21.33	22.04	No Interference	No Interference
HLRS 2025-01-300				30				70	4	7,800	30.97	31.98	No Interference	No Interference	No Interference
HLRS 2025-02-100			R0.2	10				50	4	7,340	10.34	10.68	11.03	11.41	12.26
HLRS 2025-02-200				20				60	4	7,570	20.66	21.32	22.04	No Interference	No Interference
HLRS 2025-02-300				30				70	4	7,800	30.97	31.97	No Interference	No Interference	No Interference

HARDMAX 2 Flute Long Neck Radius End Mills

Model Number	Excellent	Outside Diameter ØD	Corner Radius CR	Effective Length ℓ_1	Length of Cut ℓ	Neck Diameter ØD ₁	Shank Taper Angle Bta	Overall Length L	Shank Diameter Ød	Price ¥	Effective Length by Inclined Angles				
											30°	1°	1°30'	2°	3°
HLRS 2025-03-100		2.5	R0.3	10	2.5	2.42	16°	50	4	7,340	10.34	10.67	11.03	11.41	12.25
HLRS 2025-03-200				20				60	4	7,570	20.65	21.32	22.03	No Interference	No Interference
HLRS 2025-03-300				30				70	4	7,800	30.97	31.97	No Interference	No Interference	No Interference
HLRS 2025-05-100			R0.5	10				50	4	7,340	10.34	10.67	11.02	11.39	12.23
HLRS 2025-05-200				20				60	4	7,570	20.65	21.31	22.02	No Interference	No Interference
HLRS 2025-05-300				30				70	4	7,800	30.97	31.96	No Interference	No Interference	No Interference
HLRS 2030-01-060		3	R0.1	6	3	2.92	16°	55	6	6,000	6.21	6.42	6.63	6.86	7.37
HLRS 2030-01-060E	○			6				55	6	6,600	6.21	6.42	6.63	6.86	7.37
HLRS 2030-01-120				12				55	6	7,800	12.40	12.81	13.23	13.69	14.72
HLRS 2030-01-160				16				60	6	9,100	16.53	17.06	17.64	18.25	19.61
HLRS 2030-01-160E	○			16				60	6	10,010	16.53	17.06	17.64	18.25	19.61
HLRS 2030-01-180				18				60	6	9,100	18.59	19.19	19.84	20.53	22.06
HLRS 2030-01-200				20				60	6	9,100	20.65	21.32	22.04	22.80	24.51
HLRS 2030-01-260				26				70	6	9,100	26.84	27.71	28.64	29.64	No Interference
HLRS 2030-01-300				30				70	6	9,100	30.97	31.97	33.04	34.19	No Interference
HLRS 2030-01-360				36				80	6	11,200	37.16	38.36	39.65	41.02	No Interference
HLRS 2030-02-060			R0.2	6				55	6	6,000	6.21	6.41	6.63	6.85	7.36
HLRS 2030-02-060E	○			6				55	6	6,600	6.21	6.41	6.63	6.85	7.36
HLRS 2030-02-120				12				55	6	7,800	12.40	12.80	13.23	13.69	14.71
HLRS 2030-02-160				16				60	6	9,100	16.53	17.06	17.63	18.24	19.60
HLRS 2030-02-160E	○			16				60	6	10,010	16.53	17.06	17.63	18.24	19.60
HLRS 2030-02-180				18				60	6	9,100	18.59	19.19	19.83	20.52	22.05
HLRS 2030-02-200				20				60	6	9,100	20.65	21.32	22.03	22.80	24.49
HLRS 2030-02-260				26				70	6	9,100	26.84	27.71	28.64	29.63	No Interference
HLRS 2030-02-300				30				70	6	9,100	30.97	31.97	33.04	34.18	No Interference
HLRS 2030-02-360				36				80	6	11,200	37.15	38.36	39.64	41.02	No Interference
HLRS 2030-03-060			R0.3	6				55	6	6,000	6.21	6.41	6.62	6.85	7.35
HLRS 2030-03-060E	○			6				55	6	6,600	6.21	6.41	6.62	6.85	7.35
HLRS 2030-03-120				12				55	6	7,800	12.40	12.80	13.22	13.68	14.70
HLRS 2030-03-160				16				60	6	9,100	16.53	17.06	17.63	18.23	19.59
HLRS 2030-03-160E	○			16				60	6	10,010	16.53	17.06	17.63	18.23	19.59
HLRS 2030-03-180				18				60	6	9,100	18.59	19.19	19.83	20.51	22.04
HLRS 2030-03-200				20				60	6	9,100	20.65	21.32	22.03	22.79	24.48
HLRS 2030-03-260				26				70	6	9,100	26.84	27.71	28.63	29.62	No Interference
HLRS 2030-03-300				30				70	6	9,100	30.96	31.97	33.03	34.18	No Interference
HLRS 2030-03-360				36				80	6	11,200	37.15	38.35	39.64	41.01	No Interference
HLRS 2030-05-060			R0.5	6				55	6	6,000	6.21	6.40	6.61	6.83	7.33
HLRS 2030-05-060E	○			6				55	6	6,600	6.21	6.40	6.61	6.83	7.33
HLRS 2030-05-120				12				55	6	7,800	12.40	12.79	13.21	13.67	14.67
HLRS 2030-05-160				16				60	6	9,100	16.52	17.05	17.62	18.22	19.57
HLRS 2030-05-160E	○			16				60	6	10,010	16.52	17.05	17.62	18.22	19.57
HLRS 2030-05-180				18				60	6	9,100	18.58	19.18	19.82	20.50	22.02
HLRS 2030-05-200				20				60	6	9,100	20.65	21.31	22.02	22.78	24.46
HLRS 2030-05-260				26				70	6	9,100	26.84	27.70	28.62	29.61	No Interference
HLRS 2030-05-300				30				70	6	9,100	30.96	31.96	33.02	34.16	No Interference
HLRS 2030-05-360				36				80	6	11,200	37.15	38.35	39.63	41.00	No Interference

HARDMAX 2 Flute Long Neck Radius End Mills

Model Number	Excellent	Outside Diameter ØD	Corner Radius CR	Effective Length ℓ_1	Length of Cut ℓ	Neck Diameter Ød ₁	Shank Taper Angle Bta	Overall Length L	Shank Diameter Ød	Price ¥	Effective Length by Inclined Angles				
											30°	1°	1°30'	2°	3°
HLRS 2030-10-080		3	R1	6	3	2.92	16°	55	6	6,000	6.20	6.39	6.59	6.80	7.28
HLRS 2030-10-080E	○			6				55	6	6,600	6.20	6.39	6.59	6.80	7.28
HLRS 2030-10-120				12				55	6	7,800	12.39	12.78	13.19	13.63	14.62
HLRS 2030-10-160				16				60	6	9,100	16.51	17.04	17.59	18.19	19.52
HLRS 2030-10-160E	○			16				60	6	10,010	16.51	17.04	17.59	18.19	19.52
HLRS 2030-10-180				18				60	6	9,100	18.58	19.17	19.79	20.47	21.96
HLRS 2030-10-200				20				60	6	9,100	20.64	21.29	21.99	22.74	24.41
HLRS 2030-10-260				26				70	6	9,100	26.83	27.68	28.60	29.57	No interference
HLRS 2030-10-300				30				70	6	9,100	30.95	31.94	33.00	34.13	No interference
HLRS 2030-10-360				36				80	6	11,200	37.14	38.33	39.60	40.96	No interference
HLRS 2040-01-080		4	R0.1	8	4	3.82	16°	65	6	7,800	8.45	8.73	9.02	9.33	10.03
HLRS 2040-01-080E	○			8				65	6	8,580	8.45	8.73	9.02	9.33	10.03
HLRS 2040-01-120				12				65	6	8,000	12.58	12.99	13.42	13.89	14.92
HLRS 2040-01-160				16				65	6	9,400	16.70	17.25	17.82	18.44	No interference
HLRS 2040-01-200				20				65	6	10,130	20.83	21.50	22.23	23.00	No interference
HLRS 2040-01-200E	○			20				65	6	11,140	20.83	21.50	22.23	23.00	No interference
HLRS 2040-01-240				24				70	6	10,130	24.95	25.76	26.63	27.55	No interference
HLRS 2040-01-320				32				80	6	10,130	33.21	34.28	35.43	No interference	No interference
HLRS 2040-01-480				48				100	6	16,200	49.71	51.32	No interference	No interference	No interference
HLRS 2040-02-080			R0.2	8				65	6	7,800	8.45	8.72	9.01	9.33	10.02
HLRS 2040-02-080E	○			8				65	6	8,580	8.45	8.72	9.01	9.33	10.02
HLRS 2040-02-120				12				65	6	8,000	12.58	12.98	13.42	13.88	14.91
HLRS 2040-02-160				16				65	6	9,400	16.70	17.24	17.82	18.44	No interference
HLRS 2040-02-200				20				65	6	10,130	20.83	21.50	22.22	22.99	No interference
HLRS 2040-02-200E	○			20				65	6	11,140	20.83	21.50	22.22	22.99	No interference
HLRS 2040-02-240				24				70	6	10,130	24.95	25.76	26.62	27.54	No interference
HLRS 2040-02-320				32				80	6	10,130	33.20	34.28	35.43	No interference	No interference
HLRS 2040-02-480				48				100	6	16,200	49.71	51.32	No interference	No interference	No interference
HLRS 2040-03-080			R0.3	8				65	6	7,800	8.45	8.72	9.01	9.32	10.01
HLRS 2040-03-080E	○			8				65	6	8,580	8.45	8.72	9.01	9.32	10.01
HLRS 2040-03-120				12				65	6	8,000	12.58	12.98	13.41	13.87	14.69
HLRS 2040-03-160				16				65	6	9,400	16.70	17.24	17.81	18.43	No interference
HLRS 2040-03-200				20				65	6	10,130	20.83	21.50	22.22	22.98	No interference
HLRS 2040-03-200E	○			20				65	6	11,140	20.83	21.50	22.22	22.98	No interference
HLRS 2040-03-240				24				70	6	10,130	24.95	25.76	26.62	27.54	No interference
HLRS 2040-03-320				32				80	6	10,130	33.20	34.28	35.42	No interference	No interference
HLRS 2040-03-480				48				100	6	16,200	49.71	51.31	No interference	No interference	No interference
HLRS 2040-05-080			R0.5	8				65	6	7,800	8.45	8.71	9.00	9.31	9.99
HLRS 2040-05-080E	○			8				65	6	8,580	8.45	8.71	9.00	9.31	9.99
HLRS 2040-05-120				12				65	6	8,000	12.57	12.97	13.40	13.86	14.88
HLRS 2040-05-160				16				65	6	9,400	16.70	17.23	17.80	18.42	No interference
HLRS 2040-05-200				20				65	6	10,130	20.82	21.49	22.21	22.97	No interference
HLRS 2040-05-200E	○			20				65	6	11,140	20.82	21.49	22.21	22.97	No interference
HLRS 2040-05-240				24				70	6	10,130	24.95	25.75	26.61	27.52	No interference
HLRS 2040-05-320				32				80	6	10,130	33.20	34.27	35.41	No interference	No interference
HLRS 2040-05-480				48				100	6	16,200	49.70	51.31	No interference	No interference	No interference

HARDMAX 2 Flute Long Neck Radius End Mills

Model Number	Excellent	Outside Diameter ØD	Corner Radius CR	Effective Length ℓ_1	Length of Cut ℓ	Neck Diameter Ød ₁	Shank Taper Angle Bta	Overall Length L	Shank Diameter Ød	Price ¥	Effective Length by Inclined Angles				
											30°	1°	1°30'	2°	3°
HLRS 2040-10-080		4	R1	8	4	3.82	16°	65	6	7,800	8.44	8.70	8.98	9.27	9.93
HLRS 2040-10-080E	○			8				65	6	8,580	8.44	8.70	8.98	9.27	9.93
HLRS 2040-10-120				12				65	6	8,000	12.56	12.96	13.38	13.83	14.83
HLRS 2040-10-160				16				65	6	9,400	16.69	17.22	17.78	18.38	19.72
HLRS 2040-10-200				20				65	6	10,130	20.82	21.48	22.18	22.94	No interference
HLRS 2040-10-200E	○			20				65	6	11,140	20.82	21.48	22.18	22.94	No interference
HLRS 2040-10-240				24				70	6	10,130	24.94	25.74	26.58	27.49	No interference
HLRS 2040-10-320				32				80	6	10,130	33.19	34.25	35.39	No interference	No interference
HLRS 2040-10-480				48				100	6	16,200	49.69	51.29	No interference	No interference	No interference
HLRS 2050-02-200		5	R0.2	20	5	4.82	16°	70	6	12,900	20.83	21.50	No interference	No interference	No interference
HLRS 2050-02-400				40				90	6	16,200	41.46	No interference	No interference	No interference	No interference
HLRS 2050-03-200			R0.3	20				70	6	12,900	20.83	21.50	No interference	No interference	No interference
HLRS 2050-03-400				40				90	6	16,200	41.45	No interference	No interference	No interference	No interference
HLRS 2050-05-200			R0.5	20				70	6	12,900	20.82	21.49	No interference	No interference	No interference
HLRS 2050-05-400				40				90	6	16,200	41.45	No interference	No interference	No interference	No interference
HLRS 2050-10-200			R1	20				70	6	12,900	20.82	21.48	No interference	No interference	No interference
HLRS 2050-10-400				40				90	6	16,200	41.44	No interference	No interference	No interference	No interference
HLRS 2080-01-120		6	R0.1	12	6	5.82	—	65	6	12,900	No interference	No interference	No interference	No interference	No interference
HLRS 2080-01-120E	○			12				65	6	14,190	No interference	No interference	No interference	No interference	No interference
HLRS 2080-01-200				20				70	6	12,900	No interference	No interference	No interference	No interference	No interference
HLRS 2080-01-300				30				100	6	16,700	No interference	No interference	No interference	No interference	No interference
HLRS 2080-01-300E	○			30				100	6	18,370	No interference	No interference	No interference	No interference	No interference
HLRS 2080-01-600				60				120	6	20,300	No interference	No interference	No interference	No interference	No interference
HLRS 2080-02-120			R0.2	12				65	6	12,900	No interference	No interference	No interference	No interference	No interference
HLRS 2080-02-120E	○			12				65	6	14,190	No interference	No interference	No interference	No interference	No interference
HLRS 2080-02-200				20				70	6	12,900	No interference	No interference	No interference	No interference	No interference
HLRS 2080-02-300				30				100	6	16,700	No interference	No interference	No interference	No interference	No interference
HLRS 2080-02-300E	○			30				100	6	18,370	No interference	No interference	No interference	No interference	No interference
HLRS 2080-02-600				60				120	6	20,300	No interference	No interference	No interference	No interference	No interference
HLRS 2080-03-120			R0.3	12				65	6	12,900	No interference	No interference	No interference	No interference	No interference
HLRS 2080-03-120E	○			12				65	6	14,190	No interference	No interference	No interference	No interference	No interference
HLRS 2080-03-200				20				70	6	12,900	No interference	No interference	No interference	No interference	No interference
HLRS 2080-03-300				30				100	6	16,700	No interference	No interference	No interference	No interference	No interference
HLRS 2080-03-300E	○			30				100	6	18,370	No interference	No interference	No interference	No interference	No interference
HLRS 2080-03-600				60				120	6	20,300	No interference	No interference	No interference	No interference	No interference
HLRS 2080-05-120			R0.5	12				65	6	12,900	No interference	No interference	No interference	No interference	No interference
HLRS 2080-05-120E	○			12				65	6	14,190	No interference	No interference	No interference	No interference	No interference
HLRS 2080-05-200				20				70	6	12,900	No interference	No interference	No interference	No interference	No interference
HLRS 2080-05-300				30				100	6	16,700	No interference	No interference	No interference	No interference	No interference
HLRS 2080-05-300E	○			30				100	6	18,370	No interference	No interference	No interference	No interference	No interference
HLRS 2080-05-600				60				120	6	20,300	No interference	No interference	No interference	No interference	No interference
HLRS 2080-10-120			R1	12				65	6	12,900	No interference	No interference	No interference	No interference	No interference
HLRS 2080-10-120E	○			12				65	6	14,190	No interference	No interference	No interference	No interference	No interference
HLRS 2080-10-200				20				70	6	12,900	No interference	No interference	No interference	No interference	No interference
HLRS 2080-10-300				30				100	6	16,700	No interference	No interference	No interference	No interference	No interference
HLRS 2080-10-300E	○			30				100	6	18,370	No interference	No interference	No interference	No interference	No interference
HLRS 2080-10-600				60				120	6	20,300	No interference	No interference	No interference	No interference	No interference

Milling Conditions for HLRS (2 Flutes)

WORK MATERIAL			Copper / OFC / TPC				PREHARDENED STEELS / HARDENED STEELS NAK / SKD (30~45HRC)				HARDENED STEELS SKD / SKT (45~55HRC)				HARDENED STEELS SKD / SKT (55~65HRC)			
Model Number	Outside Diameter (mm)	Effective Length (mm)	Spindle Speed (min ⁻¹)	Feed Rate (mm/min)	a _p Axial Depth (mm)	a _e Radial Depth (mm)	Spindle Speed (min ⁻¹)	Feed Rate (mm/min)	a _p Axial Depth (mm)	a _e Radial Depth (mm)	Spindle Speed (min ⁻¹)	Feed Rate (mm/min)	a _p Axial Depth (mm)	a _e Radial Depth (mm)	Spindle Speed (min ⁻¹)	Feed Rate (mm/min)	a _p Axial Depth (mm)	a _e Radial Depth (mm)
2002	0.2	0.5	55,000	230	0.027	0.02	55,000	230	0.006	0.02	44,800	236	0.005	0.02	19,000	30	0.002	0.015
		1	55,000	200	0.027	0.02	55,000	200	0.006	0.02	35,000	150	0.004	0.02	15,000	25	0.0015	0.015
		1.5	55,000	180	0.017	0.01	55,000	180	0.005	0.01	27,000	100	0.003	0.01	12,000	20	0.001	0.007
		2	55,000	170	0.007	0.005	55,000	170	0.003	0.005	20,000	60	0.002	0.005	10,500	15	0.001	0.003
2003	0.3	1	60,000	500	0.03	0.02	60,000	500	0.007	0.02	35,000	350	0.005	0.02	22,000	35	0.004	0.015
		1.5	60,000	470	0.03	0.02	60,000	470	0.007	0.02	35,000	310	0.005	0.018	22,000	33	0.004	0.015
		2	60,000	400	0.03	0.02	60,000	400	0.007	0.02	33,200	250	0.005	0.015	20,000	32	0.004	0.015
		2.5	57,000	330	0.03	0.017	57,000	330	0.007	0.017	30,000	180	0.003	0.012	18,000	30	0.002	0.012
		3	52,000	220	0.03	0.015	52,000	220	0.006	0.015	25,000	80	0.003	0.01	15,000	20	0.002	0.01
2004	0.4	1	50,900	610	0.048	0.063	50,900	510	0.013	0.072	40,700	370	0.011	0.072	24,200	40	0.004	0.072
		1.5	45,200	580	0.045	0.063	45,200	480	0.012	0.054	36,200	360	0.01	0.054	21,500	38	0.004	0.054
		2	40,400	540	0.042	0.054	40,400	450	0.011	0.045	32,300	330	0.009	0.045	19,200	35	0.004	0.045
		3	33,900	460	0.027	0.054	33,900	390	0.008	0.027	27,100	280	0.007	0.027	16,100	30	0.003	0.027
		4	30,000	220	0.01	0.045	30,000	340	0.006	0.014	24,000	250	0.005	0.014	14,300	27	0.002	0.014
2005	0.5	1	49,200	1,370	0.081	0.117	49,200	1,140	0.034	0.122	40,000	860	0.03	0.122	24,800	94	0.013	0.122
		2	39,900	1,000	0.075	0.108	39,900	830	0.029	0.117	32,500	630	0.026	0.117	20,100	68	0.011	0.117
		3	31,900	770	0.057	0.09	31,900	640	0.023	0.113	26,000	480	0.02	0.113	16,100	52	0.008	0.113
		4	29,100	660	0.039	0.072	29,100	550	0.016	0.108	23,700	410	0.014	0.108	14,600	45	0.006	0.108
		5	26,400	570	0.027	0.045	26,400	470	0.011	0.099	21,500	360	0.01	0.099	13,300	39	0.004	0.099
		6	24,200	480	0.021	0.018	24,200	400	0.007	0.09	19,700	300	0.006	0.09	12,200	33	0.003	0.09
2006	0.6	2	28,600	610	0.114	0.162	28,600	510	0.01	0.219	23,700	390	0.01	0.219	15,200	43	0.004	0.219
		3	23,800	480	0.09	0.135	23,800	400	0.008	0.108	19,700	300	0.007	0.108	12,600	33	0.003	0.108
		4	20,400	400	0.063	0.108	20,400	330	0.005	0.104	16,800	250	0.005	0.104	10,800	28	0.002	0.104
		6	16,800	300	0.036	0.045	16,800	250	0.003	0.099	13,900	190	0.003	0.099	8,900	21	0.001	0.099
		8	14,600	240	0.021	0.027	14,600	200	0.002	0.072	12,100	150	0.002	0.072	7,700	16	0.001	0.072
2007	0.7	4	18,400	480	0.087	0.162	18,400	400	0.008	0.117	15,500	310	0.008	0.117	10,200	35	0.004	0.117
		6	15,400	360	0.051	0.108	15,400	300	0.005	0.108	13,000	230	0.005	0.108	8,600	26	0.002	0.108
2008	0.8	4	17,500	540	0.132	0.198	17,500	450	0.014	0.117	15,000	360	0.015	0.117	10,200	41	0.007	0.117
		6	14,600	410	0.075	0.144	14,600	340	0.008	0.108	12,500	270	0.008	0.108	8,500	30	0.004	0.108
		8	12,800	310	0.03	0.1	12,800	270	0.005	0.09	11,000	185	0.004	0.09	7,600	20	0.002	0.09
2010	1	2	17,600	1,100	0.21	0.45	17,600	920	0.035	0.27	15,300	750	0.04	0.27	10,900	89	0.02	0.27
		3	15,500	1,050	0.205	0.425	15,500	870	0.031	0.27	13,200	720	0.037	0.27	9,400	86	0.018	0.27
		4	13,800	980	0.201	0.405	13,800	820	0.03	0.27	12,000	670	0.035	0.27	8,500	80	0.017	0.27
		5	12,500	900	0.16	0.4	12,500	720	0.025	0.24	11,000	600	0.03	0.24	7,800	72	0.015	0.24
		6	11,300	790	0.117	0.387	11,300	650	0.021	0.216	9,800	540	0.024	0.216	7,000	64	0.012	0.216
		8	9,800	590	0.072	0.36	9,800	490	0.016	0.189	8,500	400	0.018	0.189	6,100	48	0.009	0.189
		10	8,800	390	0.048	0.315	8,800	320	0.011	0.126	7,600	270	0.013	0.126	5,400	32	0.006	0.126
		12	8,100	260	0.033	0.27	8,100	210	0.008	0.072	7,000	180	0.009	0.072	5,000	21	0.004	0.072
		16	7,000	230	0.018	0.225	7,000	190	0.004	0.027	6,100	160	0.005	0.027	4,300	19	0.002	0.027
		20	6,300	160	0.015	0.18	6,300	130	0.003	0.018	5,500	110	0.003	0.018	3,900	13	0.001	0.018
2012	1.2	6	9,400	700	0.186	0.468	9,400	580	0.018	0.09	8,400	490	0.022	0.09	6,200	60	0.011	0.09
		12	6,800	440	0.054	0.405	6,800	370	0.007	0.072	6,100	310	0.008	0.072	4,500	38	0.004	0.072
		20	5,400	250	0.021	0.24	5,400	210	0.003	0.018	4,800	180	0.003	0.018	3,500	22	0.002	0.018
2015	1.5	4	13,200	1,310	0.3	0.675	13,200	1,090	0.045	0.45	12,000	950	0.06	0.45	9,200	124	0.033	0.45
		6	10,600	1,240	0.282	0.63	10,600	1,030	0.041	0.405	9,700	900	0.055	0.405	7,400	117	0.03	0.405
		8	9,300	1,050	0.204	0.612	9,300	870	0.034	0.315	8,500	760	0.045	0.315	6,500	99	0.025	0.315
		10	8,500	900	0.15	0.567	8,500	750	0.032	0.288	7,800	650	0.042	0.288	6,000	85	0.023	0.288
		12	7,800	800	0.114	0.54	7,800	670	0.029	0.27	7,100	580	0.038	0.27	5,400	76	0.021	0.27
		16	6,800	620	0.066	0.45	6,800	510	0.015	0.18	6,200	450	0.02	0.18	4,700	58	0.011	0.18
		20	6,000	490	0.042	0.36	6,000	410	0.005	0.108	5,500	360	0.006	0.108	4,200	46	0.003	0.108

Milling Conditions for HLRS (2 Flutes)

WORK MATERIAL			Copper / OFC / TPC				PREHARDENED STEELS / HARDENED STEELS NAK / SKD (30~45HRC)				HARDENED STEELS SKD / SKT (45~55HRC)				HARDENED STEELS SKD / SKT (55~65HRC)			
Model Number	Outside Diameter (mm)	Effective Length (mm)	Spindle Speed (min ⁻¹)	Feed Rate (mm/min)	a _p Axial Depth (mm)	a _e Radial Depth (mm)	Spindle Speed (min ⁻¹)	Feed Rate (mm/min)	a _p Axial Depth (mm)	a _e Radial Depth (mm)	Spindle Speed (min ⁻¹)	Feed Rate (mm/min)	a _p Axial Depth (mm)	a _e Radial Depth (mm)	Spindle Speed (min ⁻¹)	Feed Rate (mm/min)	a _p Axial Depth (mm)	a _e Radial Depth (mm)
2020	2	4	15,300	1,500	0.33	0.9	15,300	1,250	0.046	0.9	14,300	1,130	0.065	0.9	11,500	162	0.039	0.9
		6	12,800	1,220	0.321	0.855	12,800	1,020	0.043	0.81	12,000	930	0.06	0.81	9,700	133	0.036	0.81
		8	11,200	1,120	0.267	0.81	11,200	930	0.039	0.72	10,400	850	0.055	0.72	8,400	121	0.033	0.72
		10	10,000	1,050	0.225	0.765	10,000	870	0.033	0.585	9,300	790	0.047	0.585	7,600	113	0.028	0.585
		12	9,100	980	0.186	0.72	9,100	820	0.031	0.45	8,500	740	0.044	0.45	6,900	107	0.026	0.45
		16	7,800	830	0.132	0.702	7,800	690	0.028	0.315	7,300	630	0.039	0.315	5,900	90	0.023	0.315
		20	7,000	770	0.093	0.666	7,000	640	0.017	0.198	6,600	580	0.024	0.198	5,300	84	0.014	0.198
		26	6,200	700	0.06	0.54	6,200	580	0.006	0.144	5,800	530	0.008	0.144	4,600	75	0.005	0.144
2025	2.5	30	6,000	670	0.05	0.45	6,000	550	0.005	0.135	5,500	500	0.005	0.135	4,400	70	0.002	0.135
		10	10,500	1,220	0.339	0.855	10,500	1,020	0.052	0.54	10,000	960	0.075	0.54	8,400	154	0.048	0.54
		20	7,800	720	0.165	0.756	7,800	600	0.024	0.225	7,500	560	0.035	0.225	6,300	91	0.022	0.225
2030	3	30	6,300	540	0.069	0.63	6,300	450	0.011	0.18	6,000	420	0.016	0.18	5,000	67	0.01	0.18
		6	14,000	2,700	0.5	0.9	14,000	1,510	0.15	0.72	13,300	1,140	0.15	0.72	12,000	270	0.1	0.72
		12	10,500	1,600	0.39	0.85	10,500	1,150	0.105	0.67	10,000	890	0.105	0.67	9,000	200	0.075	0.67
		16	9,200	1,160	0.321	0.81	9,200	960	0.081	0.63	8,800	730	0.081	0.63	7,900	173	0.054	0.63
		18	8,800	1,100	0.29	0.79	8,800	900	0.078	0.6	8,300	700	0.078	0.6	7,500	160	0.048	0.6
		20	8,400	1,050	0.26	0.78	8,400	880	0.073	0.58	7,900	680	0.073	0.58	7,100	150	0.044	0.58
		26	7,500	980	0.18	0.72	7,500	820	0.065	0.495	7,100	620	0.065	0.495	6,400	146	0.043	0.495
		30	7,000	870	0.14	0.69	7,000	720	0.05	0.38	6,500	560	0.05	0.38	6,000	118	0.029	0.36
2040	4	36	6,400	710	0.09	0.63	6,400	590	0.022	0.18	6,100	440	0.022	0.18	5,500	105	0.014	0.18
		8	10,200	1,340	0.42	1.62	10,200	1,110	0.095	1.35	8,500	970	0.14	1.35	7,300	223	0.101	1.35
		12	8,900	1,300	0.41	1.56	8,900	1,080	0.083	1.15	7,600	950	0.12	1.15	6,400	215	0.085	1.15
		16	7,900	1,250	0.4	1.5	7,900	1,030	0.065	1	6,600	910	0.1	1	5,600	205	0.065	1
		20	6,900	1,190	0.384	1.44	6,900	990	0.054	0.9	5,800	860	0.08	0.9	4,900	198	0.058	0.9
		24	6,200	1,100	0.31	1.38	6,200	900	0.043	0.8	5,200	770	0.065	0.8	4,500	175	0.043	0.8
2050	5	32	5,500	860	0.189	1.26	5,500	720	0.027	0.648	4,600	630	0.04	0.648	3,900	144	0.029	0.648
		48	4,600	430	0.093	1.08	4,600	360	0.007	0.315	3,900	310	0.01	0.315	3,300	72	0.007	0.315
		20	6,700	1,780	0.606	1.98	6,700	1,480	0.092	1.17	4,800	990	0.13	1.17	4,000	297	0.096	1.17
2060	6	40	4,600	850	0.297	1.53	4,600	710	0.046	0.9	3,300	470	0.065	0.9	2,800	143	0.048	0.9
		12	8,000	1,800	0.6	2.25	8,000	1,620	0.5	1.35	4,700	1,360	0.2	1.35	4,000	540	0.15	1.35
		20	5,800	1,350	0.58	2.12	5,800	1,180	0.46	1.31	3,500	1,000	0.18	1.31	3,000	380	0.14	1.31
		30	4,500	1,060	0.546	1.98	4,500	880	0.396	1.26	2,600	740	0.158	1.26	2,200	294	0.119	1.26
2060	6	60	2,800	530	0.156	1.62	2,800	440	0.113	0.99	1,600	370	0.045	0.99	1,400	147	0.034	0.99



Note:

- Decrease both spindle speed and feed rate proportionally when the milling parameters exceed the machine's maximum spindle speed
- Every coolant offers stable milling.
- Recommend oil coolant for Stainless Steels and Heat Resistant Alloys.
- Recommend wet coolant for Copper.



Advisory for Safe Use of UNIMAX End Mills

Correct application and operation is strongly advised to avoid clogging, abrasion, etc, that could cause serious accidents or injuries. Ignition or sparks generated during milling could lead to fire or extreme damage to the work piece. End Mills are made with very sharp cutting edges and must be handled with extra care.

- * Never touch the cutting edge with your bare hands, as this could cause serious injury. Special caution is required when opening the package.
- * Dropping the tool could cause breakage or flying debris, leading to serious injury.
- * During milling, unexpected impact or shock on the tool could cause breakage or flying debris. Ensure to use protective items such as safety glasses and a face guard.
- * For best results, fine parameter adjustment may be required, depending on the materials; milling shape and strategy; machine rigidity and spindle capability.
- * Use a machine that has high rigidity and generates a low level of vibration.
- * Do not use flammable cutting oils.

Advisory for regrinding UNIMAX End Mills

- * Never regrind the tool without wearing safety glasses and a face guard.



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